Raktsetu

eRaktKosh Blood Lakehouse: National Blood Availability & Shortage Intelligence

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Submitted to

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Declaration

We, Radhey Gajera (25MCA057) and Shruti Gajipara (25MCA058), students of Master of Computer Applications (MCA) at Smt. Chandaben Mohanbhai Patel Institute of Computer Applications (CMPICA), CHARUSAT, declare that the project report titled “Raktsetu: eRaktKosh Blood Lakehouse” is our own work. It was carried out under the guidance of Ms. Konica Soni.

This work has not been submitted to any other university or institute for any degree or diploma. All data sources, tools and references used in this project are acknowledged in this report.

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Acknowledgement

Knowledge in itself is a continuous process. At this moment of our substantial enhancement, we rarely find words to express our gratitude towards those who were constantly involved with us.

The completion of any inter-disciplinary project depends upon coordination, cooperation and the combined efforts of several resources of knowledge, creativity, skill, energy and time. With the work now accomplished, we take this opportunity to acknowledge, in these lines, everyone who contributed to it.

We would like to thank our project guide Ms. Konica Soni, I/C Principal **Dr. Dharmendra Patel** and I/C Dean **Dr. Sanskruti Patel** who advised and gave us moral support throughout the duration of our project. Without their constant encouragement we could not have achieved what we have.

We also thank the Ministry of Health and Family Welfare (MoHFW) for publishing blood stock data openly through the eRaktKosh portal, and the open-source communities behind DuckDB, dbt, FastAPI, Leaflet and OpenStreetMap, whose work made this project possible.

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With Sincere Regards,

Radhey Gajera (25MCA057)
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Abstract

When a patient needs blood urgently, families often phone one blood bank after another. India's eRaktKosh portal already publishes blood stock for thousands of blood centres, but it is searched one district at a time, gives no help with blood-group compatibility and does not show how old each record is.

Raktsetu solves this by building a data lakehouse on top of eRaktKosh. An async scraper collects stock for every district and saves it as immutable Parquet files. DuckDB and dbt clean the data in Bronze, Silver and Gold layers, keep a daily history and run 37 data-quality tests. Only data that passes these checks is published to a read-only database, which a FastAPI REST API serves to a responsive website.

The website offers a Smart Emergency Mode that lists compatible donor groups (including plasma and the rare Bombay group) and ranks nearby centres by freshness, trust, units and distance. It also provides a blood finder with a live map, 7-day trends, a Trust Score, rare-blood and district-shortage radars, CSV export, a donor eligibility checker and a data-health dashboard. It works on any phone and offline.

The system covers 4,716 blood centres in 35 States/UTs and 706 districts. It is supported by 98 automated code tests, 37 data tests and 146 end-to-end browser checks, all passing.

Keywords: *blood availability, eRaktKosh, data lakehouse, DuckDB, dbt, FastAPI, Progressive Web App, emergency healthcare*

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PROJECT PROFILE

Project Profile

Project Name: Raktsetu (eRaktKosh Blood Lakehouse)

Type of Application: Web Application (Progressive Web App) with a Data Engineering Pipeline

Project Description: Raktsetu turns the blood-stock data published by the Government of India's eRaktKosh portal into a fast, honest emergency finder. A medallion data lakehouse collects nationwide stock from every district, cleans and tests it, keeps a daily history, and publishes it to a responsive website. Patients' families can find compatible blood near them in seconds, see how fresh each record is, call or navigate to the blood centre, and officials can spot district-level shortages and export them as reports.

Data Source: eRaktKosh (Ministry of Health and Family Welfare, Govt. of India), public portal

Coverage: 4,716 blood centres across 35 States/UTs and 706 districts

Team Size: 2

Frontend: HTML5, CSS3, Vanilla JavaScript, Leaflet.js (maps), Chart.js (charts)

Backend: Python 3.12, FastAPI, Uvicorn

Data Pipeline: Async scraper (aiohttp), Apache Parquet, dbt Core (dbt-duckdb)

Database: DuckDB: analytical warehouse plus a read-only serving replica

Tools used: VS Code, uv, Git & GitHub, Docker, pytest, Playwright, Ruff

Internal Guide: Ms. Konica Soni, Assistant Professor, CMPICA

Project Timeline (Gantt Chart)

√ The project was planned in eight phases over sixteen weeks. Some phases overlap, for example the website was built while the API was being finished.

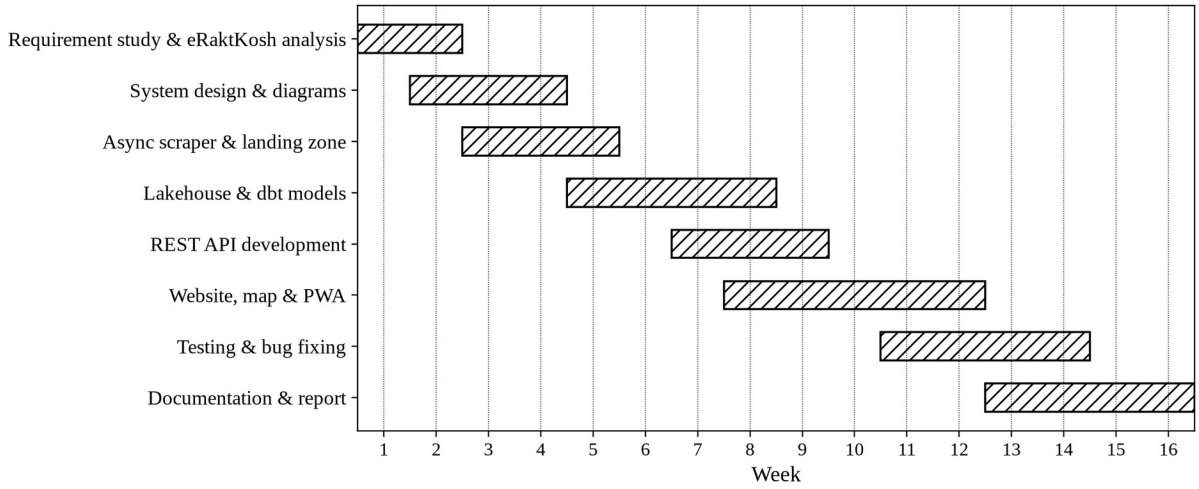


Figure 1: Project timeline (Gantt chart)

INTRODUCTION TO TOOLS

Introduction to Tools & Technology

❖ Front End:

√ HTML5 & CSS3

- **Purpose:** Structure and visual design of every screen of the website.
- **Features:** Semantic sections, CSS Grid/Flexbox layouts, media queries, design tokens.
- **Benefits:** Fully responsive from 320 px phones to 1920 px desktops without any UI framework.

√ JavaScript (ES2020)

- **Purpose:** All client-side behaviour: search, filters, modals, map and charts.
- **Features:** Fetch API, async/await, URL state sync, Geolocation API, client-side eligibility rules.
- **Benefits:** Zero build step and a small download, so the site loads fast on slow mobile networks.

√ Leaflet.js

- **Purpose:** Interactive map of blood centres.
- **Features:** Custom markers, pop-ups, fit-to-results and a 'you are here' marker.
- **Benefits:** Lightweight open-source mapping that works well on mobile devices.

√ Chart.js

- **Purpose:** National distribution chart and per-bank historical trend chart.
- **Features:** Bar and line charts with tooltips; gaps are shown for days without data.
- **Benefits:** Clear visual insight with very little code.

√ Service Worker & Web App Manifest (PWA)

- **Purpose:** Installable app with offline support.
- **Features:** Network-first caching of the app shell, libraries and last API responses.
- **Benefits:** The last known inventory stays visible during network outages.

Back End & Data Engineering

❖ Back End:

√ Python 3.12

- **Purpose:** Primary language for the API, scraper, pipeline CLI and tests.
- **Features:** Type hints, asyncio, rich standard library.
- **Benefits:** Readable, maintainable code with a large data ecosystem.

√ FastAPI & Uvicorn

- **Purpose:** REST API serving the website and third-party clients.
- **Features:** Automatic OpenAPI docs (/docs), request validation with Pydantic, CSV export endpoints.
- **Benefits:** High performance with self-documenting, type-checked endpoints.

❖ Data Engineering:

√ aiohttp (Async Scraper)

- **Purpose:** Collects blood stock for every district from eRaktKosh.
- **Features:** 10 concurrent requests, retries with exponential back-off, circuit breaker at 20% errors.
- **Benefits:** A full national crawl in minutes while staying polite to the government server.

√ Apache Parquet

- **Purpose:** Immutable landing zone for every raw scrape.
- **Features:** Columnar, compressed files partitioned by date and hour with a SHA-256 manifest.
- **Benefits:** Raw data is never lost and any run can be reloaded or audited.

√ DuckDB

- **Purpose:** Embedded analytical database (warehouse and serving copy).
- **Features:** Columnar SQL engine, reads Parquet directly, single-file database.
- **Benefits:** Analytics in milliseconds with no database server to install or manage.

√ dbt Core (dbt-duckdb)

- **Purpose:** Transforms raw data into clean Silver and Gold tables.
- **Features:** Version-controlled SQL models, incremental snapshots, 37 automated data tests, lineage docs.
- **Benefits:** Reliable, testable and documented transformations.

Key Services & Architecture

❖ Key Services:

√ eRaktKosh Public Portal

- **Purpose:** Single source of truth for blood stock in India.
- **Features:** District-wise availability per blood group and component for registered blood centres.
- **Benefits:** Official, nationwide data; every page and export credits MoHFW as the source.

√ OpenStreetMap Nominatim

- **Purpose:** Geocoding of all 706 districts to map coordinates.
- **Features:** One-time lookup at 1 request/second, cached in data/district_coords.json.
- **Benefits:** Accurate 'Near me' distances without a paid geocoding service.

√ CARTO Positron Basemap

- **Purpose:** Muted background map tiles.
- **Features:** Light, low-contrast tiles that keep blood centre markers easy to read.
- **Benefits:** Clean, professional map appearance.

❖ Architecture:

√ Medallion Lakehouse (Bronze, Silver, Gold)

- **Purpose:** Layered data design.
- **Features:** Bronze keeps raw rows, Silver cleans and parses, Gold holds analysis-ready marts.
- **Benefits:** Each layer is testable and every number can be traced back to its raw source.

√ Atomic Blue/Green Publishing

- **Purpose:** Safe release of new data to the website.
- **Features:** An anomaly gate checks the data, then the warehouse is copied to serving.duckdb atomically.
- **Benefits:** Users never see half-updated or broken data.

Development & Testing Tools

√ **uv & Ruff**

- **Purpose:** Dependency management, linting and formatting.
- **Features:** Locked, reproducible environments; fast lint and format checks.
- **Benefits:** Consistent code quality across the team.

√ **pytest**

- **Purpose:** Unit and integration testing of Python code.
- **Features:** 98 automated tests covering scraper, scoring, geocoding, CSV export and every API endpoint.
- **Benefits:** Regressions are caught before they reach users.

√ **Playwright**

- **Purpose:** Automated browser (end-to-end) testing.
- **Features:** Drives the real website on desktop and mobile screen sizes; 146 user-flow checks.
- **Benefits:** Proves that features work from the user's side, not just in code.

√ **Docker & Git**

- **Purpose:** Packaging and version control.
- **Features:** Container image with health check; Git history on GitHub.
- **Benefits:** The same build runs on any machine; all changes are tracked.

SYSTEM STUDY

Existing System

Existing System: eRaktKosh Portal (MoHFW, Govt. of India)

- √ **District search:** Users select a state, district, blood group and component to view stock.
- √ **Facility details:** Shows the blood centre name, contact details and last update time.
- √ **Government data:** Stock is entered by registered blood centres across India.
- √ **Web access:** Available as a public website with no login needed for searching.

❖ Limitations of the Existing System:

- Search works one district at a time; there is no nationwide or 'near me' view.
- No blood-group compatibility help: a family searching O- is not told which donor groups are safe.
- No indication of how fresh a record is, so months-old stock looks the same as today's.
- No history, trends or shortage view for health officials.
- No dedicated search for rare groups such as the Bombay (Oh) phenotype.
- No data export for reports and planning.

Proposed System

Raktsetu is a web platform built on a data lakehouse. It collects eRaktKosh stock for every district, cleans and validates it, keeps a daily history, and presents it through an emergency-first, mobile-friendly website. It adds clinical compatibility rules, freshness and trust indicators, distance ranking and analytics that the existing portal does not offer.

❖ Key Features

√ Smart Emergency Mode

- Enter the patient's blood group and component; the system lists compatible donor groups (NBTC rules, including plasma and Bombay rules).
- Ranks blood centres by freshness, trust, units and distance, with one-tap Call and Directions.

√ Blood Finder & Live Map

- Filter by state, district, component, blood group and name; sort by freshness, units, trust or distance.
- 'Near me' uses the device location to show the closest centres on a map.

√ Freshness, Trust Score & 7-day Trend

- Every card shows when the stock was last updated and a 0–100 Trust Score.
- A trend chart shows real daily snapshots; days without data appear as gaps, never invented values.

√ Rare Blood & District Shortage Radar

- Nationwide list of Bombay and rare-group stock.
- District-level shortage severity for each blood group, exportable as CSV.

√ Donor Eligibility Checker

- Checks age, weight, haemoglobin, recent tattoo/surgery/illness and donation gap in the browser only.
- Shows the next eligible donation date; no personal data leaves the device.

√ Data Health Dashboard & PWA

- Shows last sync time, fresh vs stale centres, dbt test results and recent pipeline runs.
- Installable, responsive and usable offline with the last cached data.

Existing vs Proposed System

Table 1: Comparison of existing and proposed system

Feature	Existing (eRaktKosh)	Proposed (Raktsetu)
Search scope	One district at a time	Whole country, any filter
Blood compatibility help	No	Yes (red cells, plasma, Bombay)
Nearest blood centre	No	Yes, using device location
Shows how fresh the data is	Partly (date only)	Yes, on every result
Trust / reliability score	No	Yes (0–100)
Stock history and trends	No	Yes (daily snapshots)
District shortage analysis	No	Yes, with severity score
Rare (Bombay) blood search	No	Yes, nationwide
Report export	No	Yes, CSV
Donor eligibility check	No	Yes, private, in browser
Works offline	No	Yes (PWA)
Open API for other apps	No	Yes (REST API)

Scope of the Proposed System

Raktsetu covers the full path from public eRaktKosh data to an emergency decision: data collection, quality control, history, analytics and a public website. It is an information service; it does not reserve or transfer blood, and users are always told to call the blood centre before travelling.

❖ **Public User Side Scope (patients' families, doctors, donors, officials)**

1. Emergency Search

Find compatible blood for a patient and call the best-ranked centres.

2. Blood Finder

Search and filter all centres, view stock per blood group, switch to the map.

3. Near Me

Use the device location to rank centres by distance.

4. Trends & Trust

View a centre's recent stock history and its Trust Score.

5. Rare Blood & Shortages

Find rare-group stock and districts that are running short.

6. Reports

Download availability and shortage data as CSV files.

7. Donor Help

Check donation eligibility and the next eligible date.

❖ **Operator (Admin) Side Scope**

1. Scrape

Run the nationwide crawl that writes an immutable Parquet file.

2. Load & Transform

Load the file into Bronze and build Silver/Gold tables with dbt.

3. Quality Gate

Run 37 data tests and the facility-count anomaly check.

4. Publish

Atomically release the new data to the website.

5. Monitor

Track sync age, stale centres and pipeline runs on the Data Health page.

Aim and Objectives

Aim: To build a reliable, data-driven web platform that converts India's public eRaktKosh blood-stock data into fast, trustworthy and clinically aware guidance for people searching for blood in an emergency, and into shortage intelligence for health officials.

Objectives

- ✓ Collect nationwide blood stock automatically and politely from every eRaktKosh district.
- ✓ Store raw data immutably and build cleaned, tested Bronze, Silver and Gold layers.
- ✓ Keep a daily snapshot history so trends can be shown honestly over time.
- ✓ Apply NBTC compatibility rules for red cells, plasma and the Bombay phenotype.
- ✓ Rank blood centres using freshness, trust, available units and distance.
- ✓ Show district-level shortages and rare-blood availability, with CSV export.
- ✓ Block publishing when data quality checks fail, so users never see broken data.
- ✓ Deliver a responsive, accessible, installable website that also works offline.
- ✓ Protect privacy: no login, no personal data stored, location never saved.

Feasibility Study

Operational Feasibility

Operational feasibility examines how well the solution fits the way people already search for blood and how easily it can be operated.

- ✓ **User Roles:** Public users search without any account. The operator runs one command (make pipeline) to refresh data.
- ✓ **User Experience (UX):** Emergency search needs only three choices and one click; results show Call and Directions buttons directly.
- ✓ **Training:** No training is needed for public users. The operator needs a short guide covering the pipeline commands.
- ✓ **Integration with Existing Process:** Raktsetu does not replace eRaktKosh; it reads its public data and adds compatibility, ranking and analytics on top.
- ✓ **Support & Maintenance:** Automated tests, the anomaly gate and the Data Health page make problems visible early.
- ✓ **Risk Factors:** If eRaktKosh changes or fails, the circuit breaker and quality gate stop the run and the last good data stays online.

Technical Feasibility

- ✓ **Technology Stack:** Python, FastAPI, DuckDB, dbt and vanilla web technologies are mature, open-source and well documented.
- ✓ **Platform Compatibility:** Runs in any modern browser on Android, iOS, Windows, macOS and Linux; installable as a PWA.
- ✓ **Performance:** DuckDB answers analytical queries over about 66,000 current stock rows in milliseconds on a laptop.
- ✓ **Reliability:** Read-only serving database and atomic publishing keep the website consistent during updates.
- ✓ **Scalability & Maintainability:** Modular code (scraper, pipeline, API, web) and dbt models can be extended independently.
- ✓ **Development Tools:** uv, Ruff, pytest, Playwright, Docker and Git provide a reproducible, tested build.

Economical Feasibility

Economic feasibility determines whether the project is cost-effective and what value it returns.

- ✓ **Development Cost:** All software used is free and open source; development was done by the two-member team.
- ✓ **Initial Investment:** No licences or paid APIs; the data source and map geocoding are free public services.
- ✓ **Operational Cost:** A single low-cost virtual server can host the API and run the daily pipeline.
- ✓ **Cost-Benefit Analysis:** Saves families time during emergencies and gives officials shortage reports without extra data entry.
- ✓ **Social Return on Investment:** Faster access to compatible blood can directly reduce delays in treatment.
- ✓ **Scalability Cost:** Costs grow slowly: DuckDB handles millions of rows on one machine before any upgrade is needed.

SYSTEM ANALYSIS

Requirement Specification (along with System Modules)

1. Data Ingestion Module:

√ Async scrape of all districts listed in the eRaktKosh metadata catalogue.

√ Retries with back-off, circuit breaker, minimum 6-hour run interval.

√ Writes an immutable Parquet file plus a manifest with SHA-256 checksum.

2. Lakehouse Transformation Module:

√ Idempotent load of Parquet into bronze.raw_inventory.

√ dbt models: Silver staging views and Gold facts, dimensions and marts.

√ Incremental daily snapshots; groups missing from the latest scrape count as 0 units.

3. Quality Gate & Publishing Module:

√ 37 dbt data tests and an 80% facility-count anomaly gate.

√ Run log in gold.pipeline_runs; atomic copy to serving.duckdb.

4. Emergency Module:

√ Compatible donor groups for red cells, plasma and Bombay phenotype.

√ Weighted ranking (freshness 35%, trust 30%, units 25%, distance 10%); radius search near the user.

5. Blood Finder & Map Module:

√ Filters, sorting, pagination, URL-shareable state, grid and map views, 'Near me'.

6. Analytics Module:

√ National distribution chart, 7-day trend, Trust Score, rare blood radar, district shortages.

7. Donor Assistance Module:

√ Client-side eligibility rules (NBTC) with next eligible date; no data sent to the server.

8. Data Health & Export Module:

√ Sync age, fresh/stale counts, dbt results, recent runs; CSV export of availability and shortages.

9. REST API Module:

√ Documented endpoints under /api/v1 with validation, OpenAPI docs and health check (/healthz).

❖ Non-Functional Requirements:

√ **Performance:** Search responses within about one second for nationwide queries.

√ **Privacy:** No login, no personal data stored; donor check and location stay in the browser.

√ **Reliability:** Website keeps serving the last good data if a pipeline run fails.

√ **Usability & Accessibility:** Responsive from 320 px, 40 px touch targets, keyboard-closable dialogs.

Hardware and Software Requirements

Server / Developer Side

Table 2: Server and developer requirements

Item	Minimum Requirement
Processor	Dual-core 2 GHz or better
RAM	4 GB (8 GB recommended)
Storage	10 GB free disk space
Operating System	Linux, Windows 10/11 or macOS
Internet	Required for scraping eRaktKosh and map tiles
Software	Python 3.12, uv, dbt-duckdb, FastAPI, Uvicorn, Git; Docker (optional)

Client (User) Side

Table 3: Client requirements

Item	Minimum Requirement
Device	Any smartphone, tablet, laptop or desktop
Screen	320 px wide or larger
Browser	Latest Chrome, Firefox, Edge or Safari
Internet	Needed for live data; the last data stays available offline
Location	Optional, only for 'Near me' (needs HTTPS or localhost)

Data Flow Diagram (Level 0)

√ The context diagram shows Raktsetu as one process and the outside entities that send data to it or receive data from it.

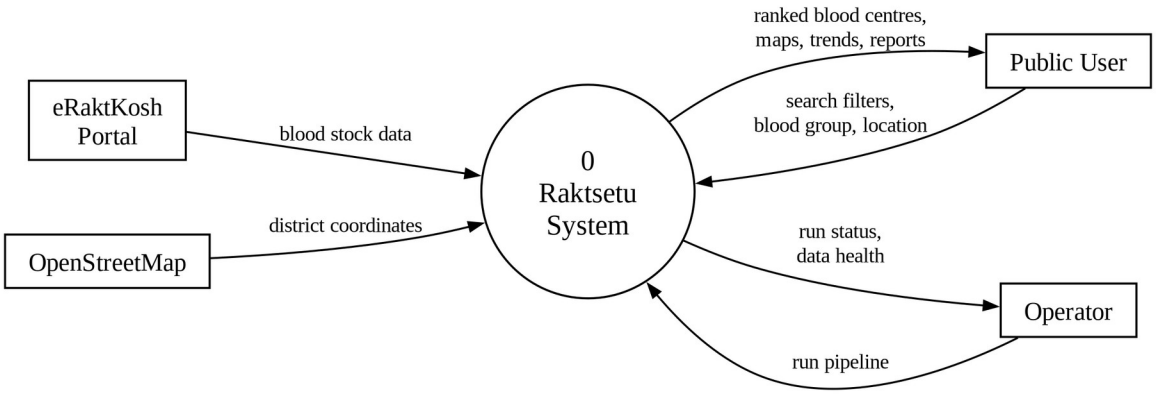


Figure 2: Data Flow Diagram (Level 0)

Data Flow Diagram (Level 1)

√ Level 1 breaks the system into six processes and shows the four data stores between them.

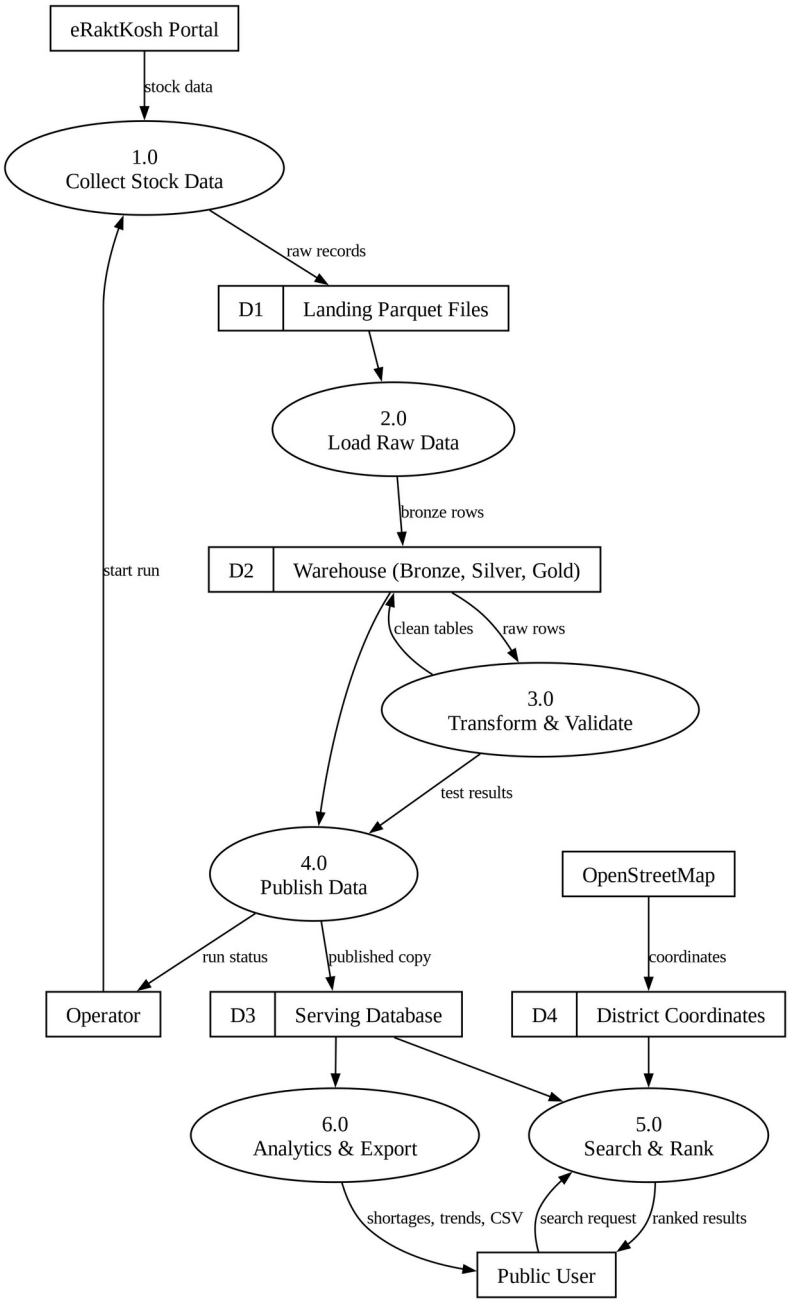


Figure 3: Data Flow Diagram (Level 1)

Use Case Diagram: Public Users

√ Four kinds of public users use the website without logging in. «include» means the step always happens; «extend» means it is optional.

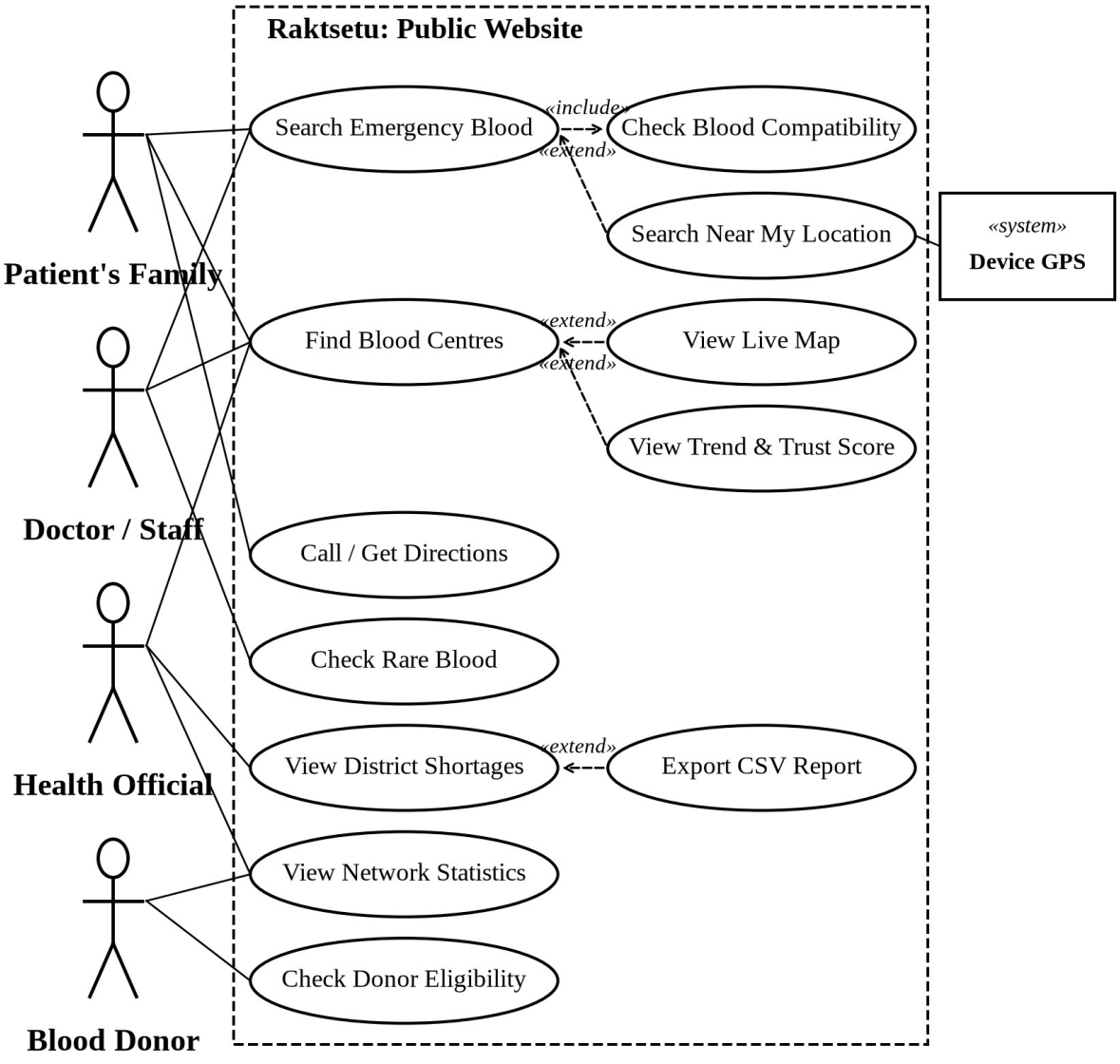


Figure 4: Use Case Diagram: Public Users

Actors of the System

Table 4: Actors of the system

Actor	Type	What they do
Patient's Family	Primary	Searches for compatible blood in an emergency and calls the centre
Doctor / Hospital Staff	Primary	Checks compatible groups, rare blood and stock trends
Health Official	Primary	Monitors district shortages and exports reports
Blood Donor	Primary	Checks donation eligibility and where blood is needed
Operator (Admin)	Primary	Runs the data pipeline and monitors data health
eRaktKosh Portal	Secondary (system)	Provides blood stock data
OpenStreetMap	Secondary (system)	Provides district coordinates and map data
Device GPS	Secondary (system)	Provides the user's location for 'Near me'

Use Case Diagram: Operator & External Systems

√ The operator runs the pipeline, which always includes scraping, loading, transforming, testing and publishing. eRaktKosh and OpenStreetMap are external systems.

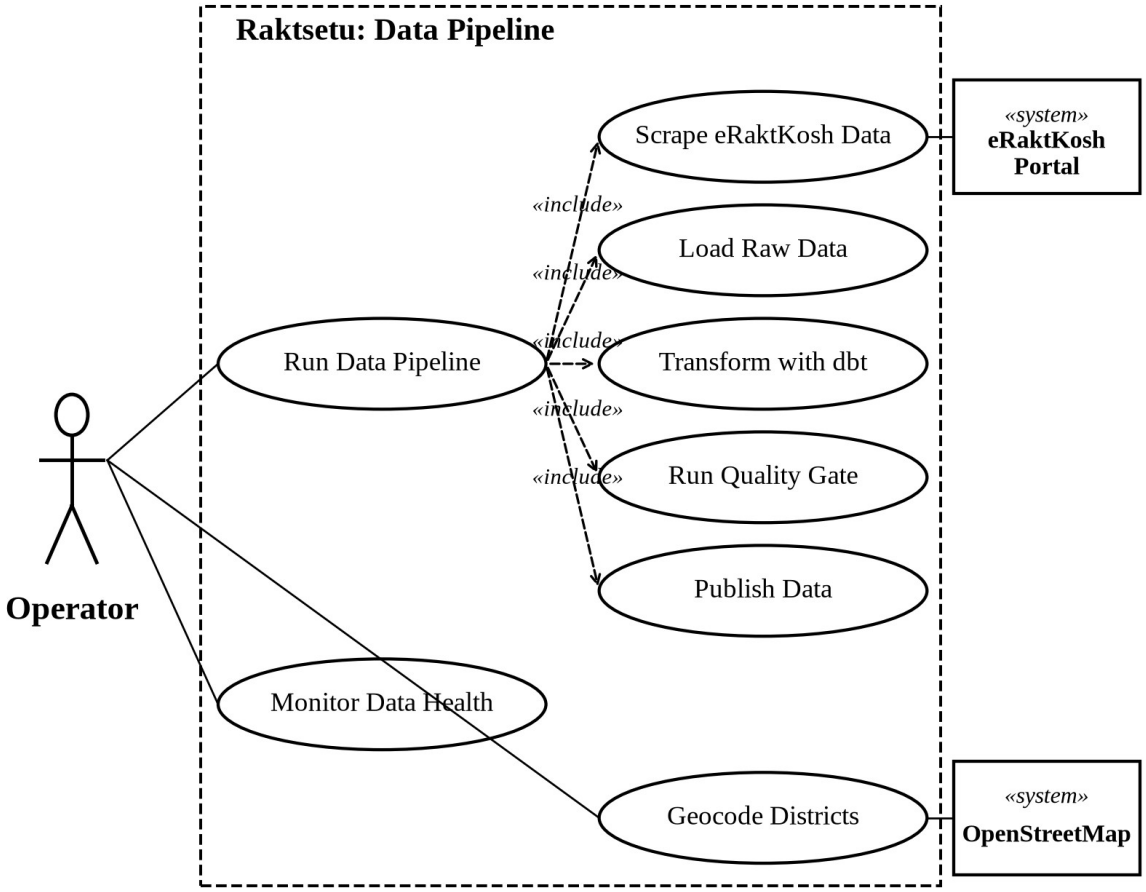


Figure 5: Use Case Diagram: Operator & External Systems

Class Diagram

√ Main classes of the system with their attributes and operations, and how they depend on each other.

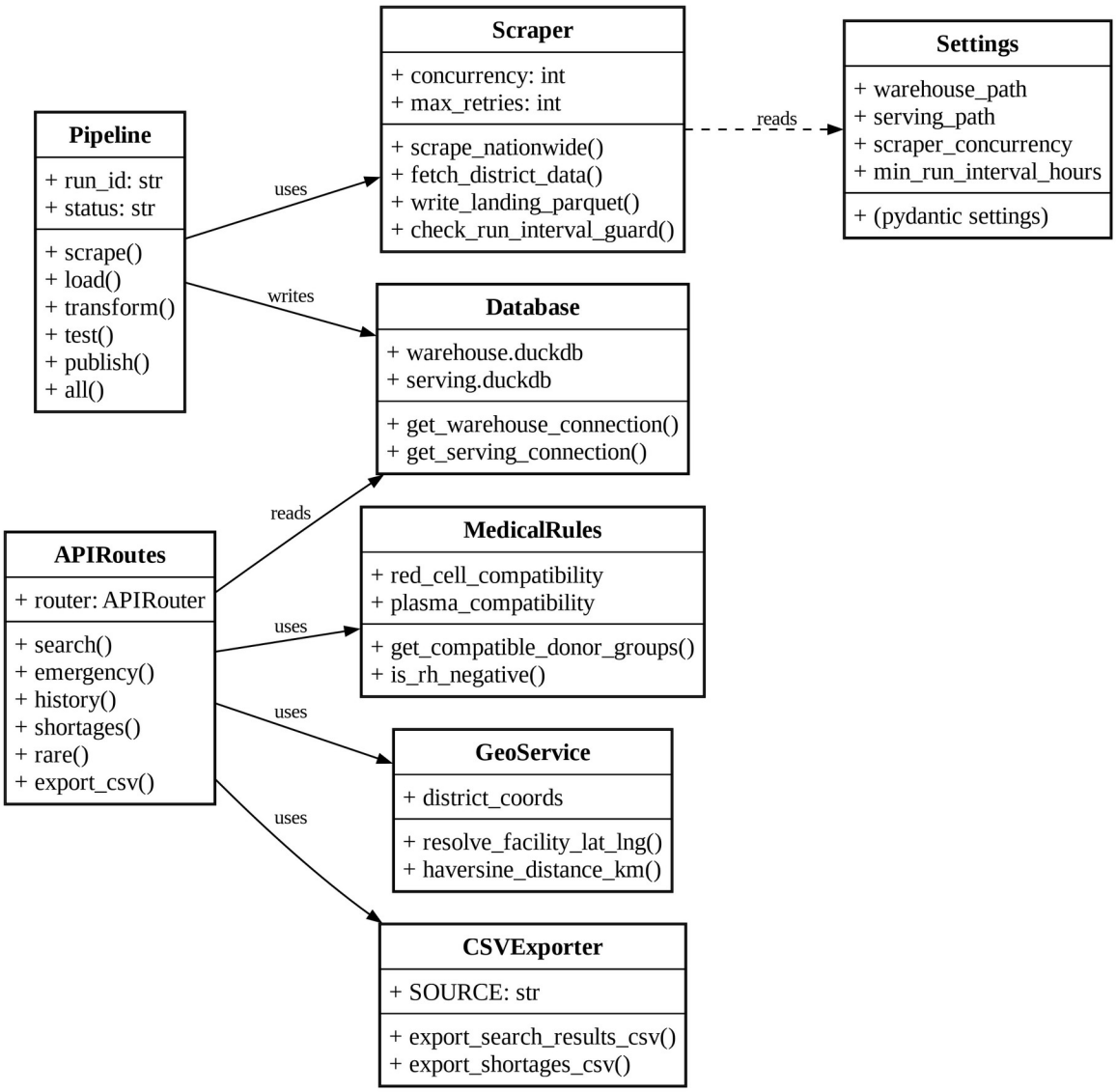


Figure 6: Class Diagram

Activity Diagram: Public User

√ How a user finds blood: emergency search or Blood Finder, then Call or Directions.

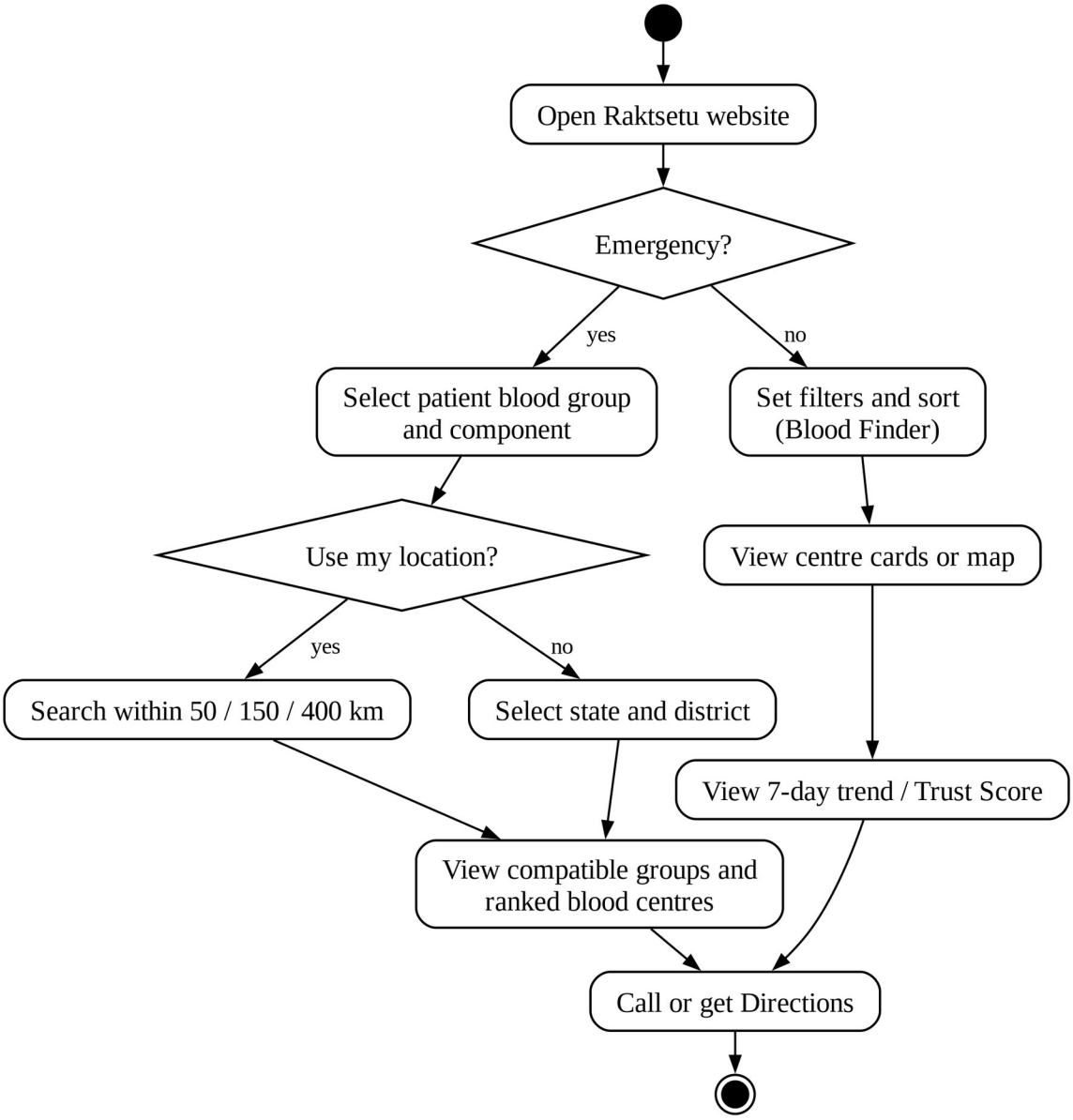


Figure 7: Activity Diagram: Public User

Activity Diagram: Data Pipeline

√ One pipeline run from scraping to publishing, including the checks that can stop it.

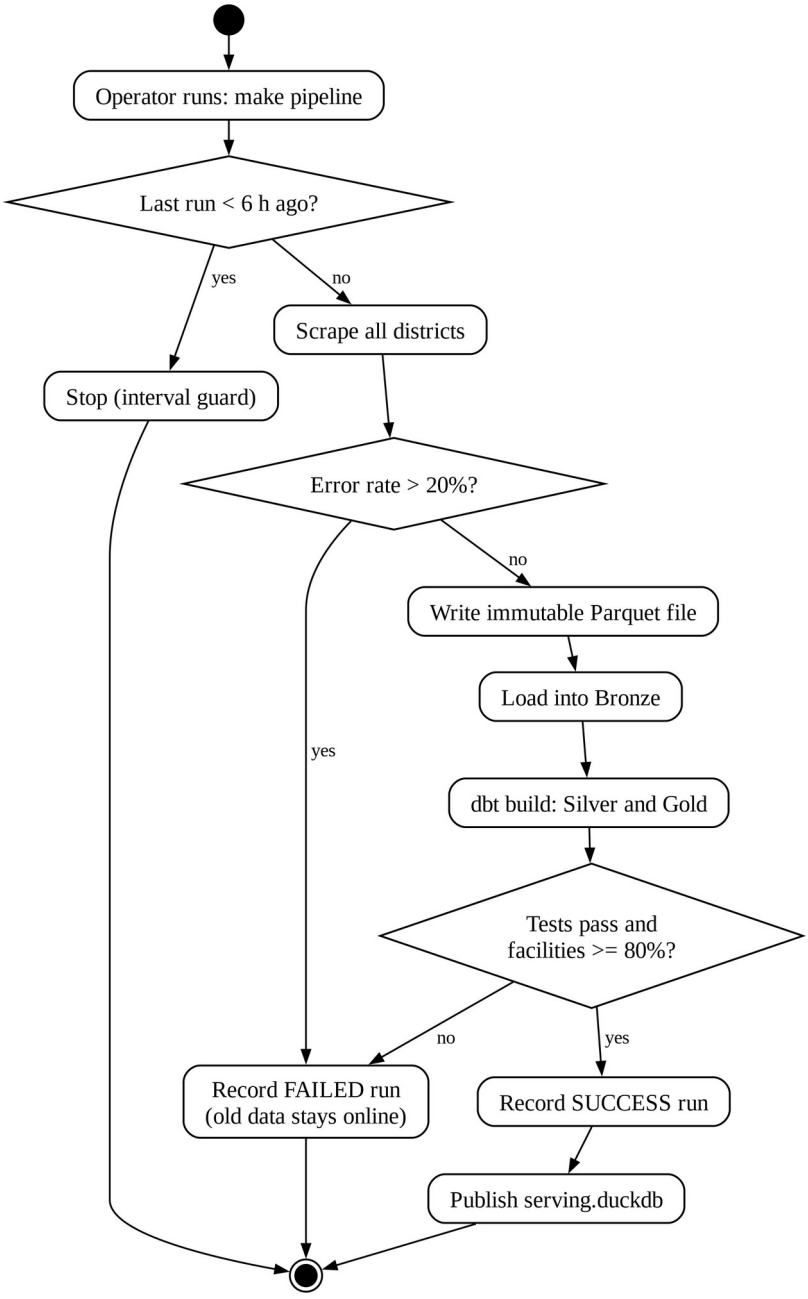


Figure 8: Activity Diagram: Data Pipeline

Sequence Diagram

√ Messages exchanged between the user, website, API, medical rules and database during an emergency search.

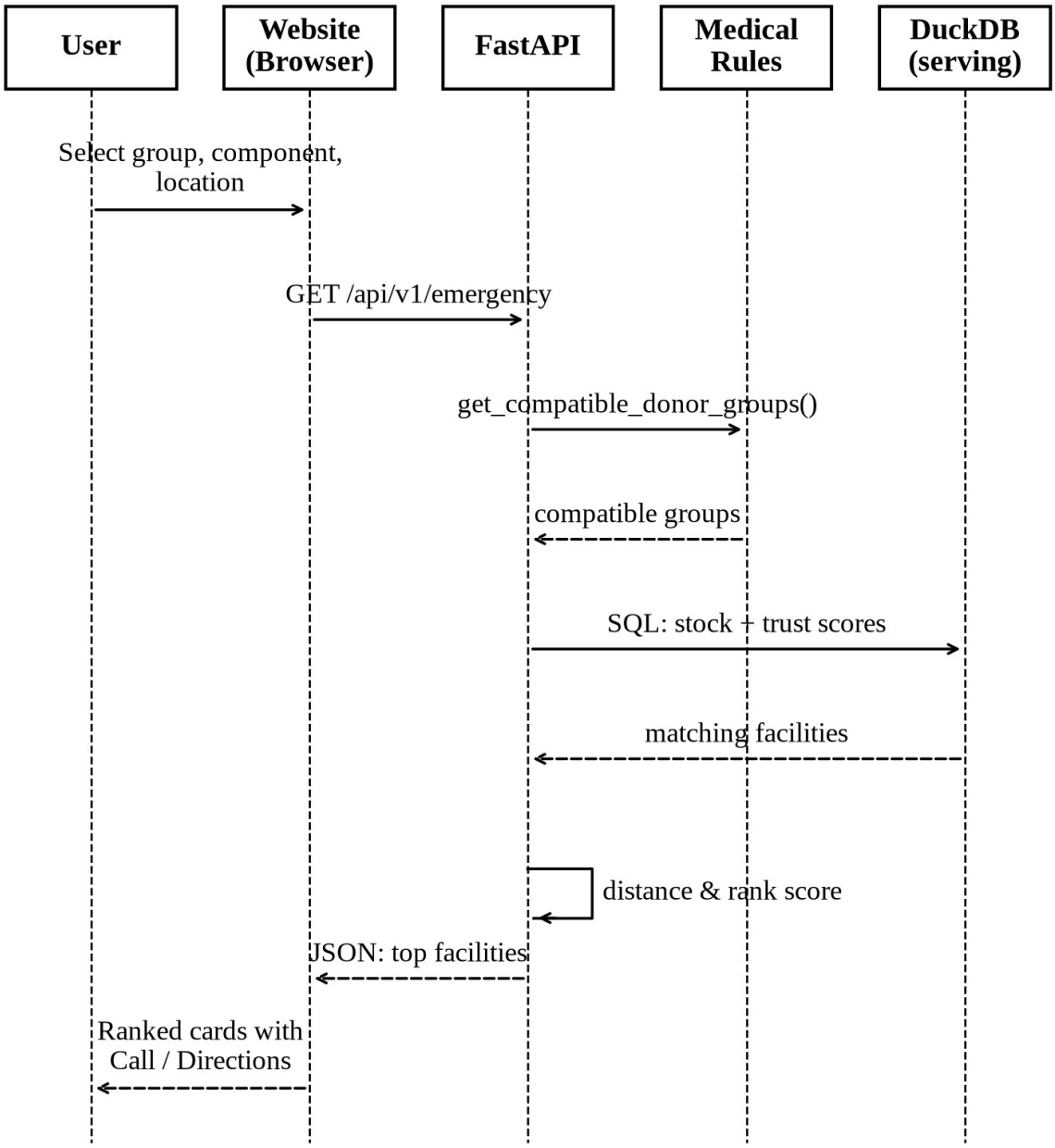


Figure 9: Sequence Diagram

System Architecture

√ End-to-end design: from the eRaktKosh portal, through the lakehouse layers and quality gate, to the website.

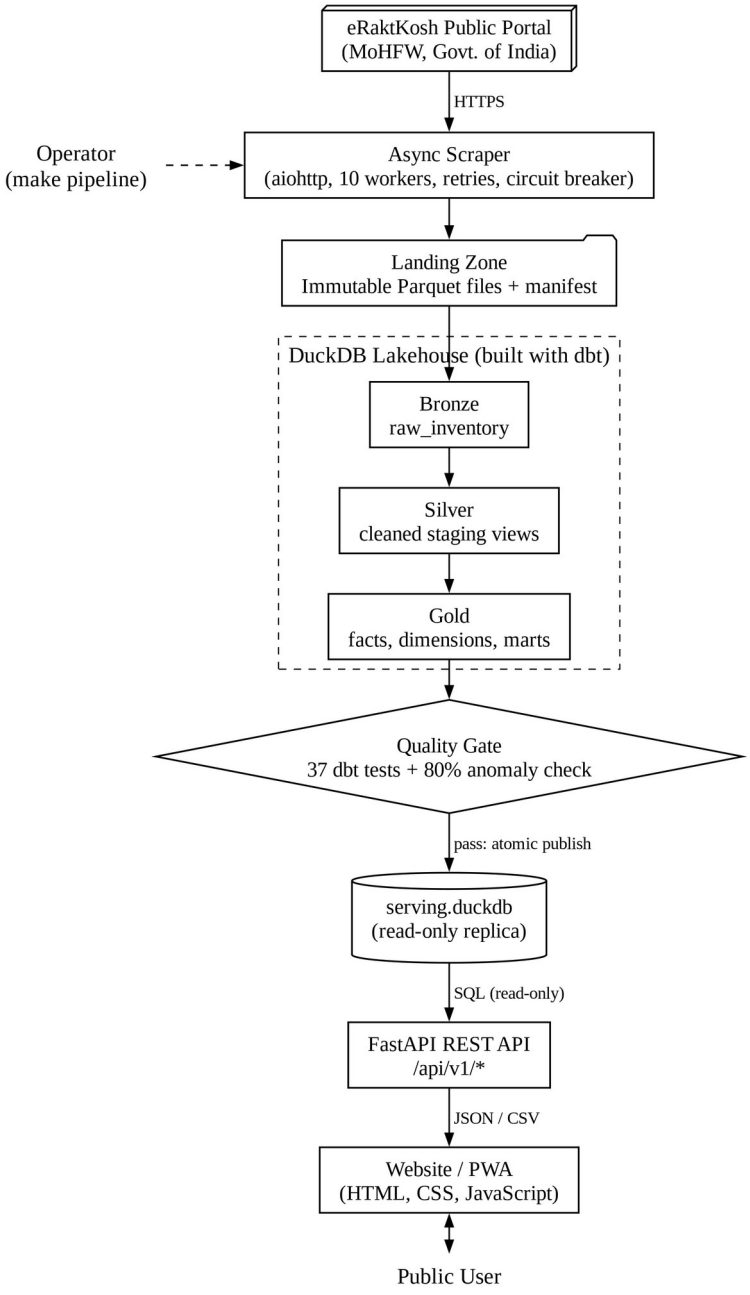


Figure 10: System Architecture

SYSTEM DESIGN

ER Diagram

√ Entities stored in the lakehouse and how they are related. PK = primary key, FK = foreign key; underlined fields together identify a row.

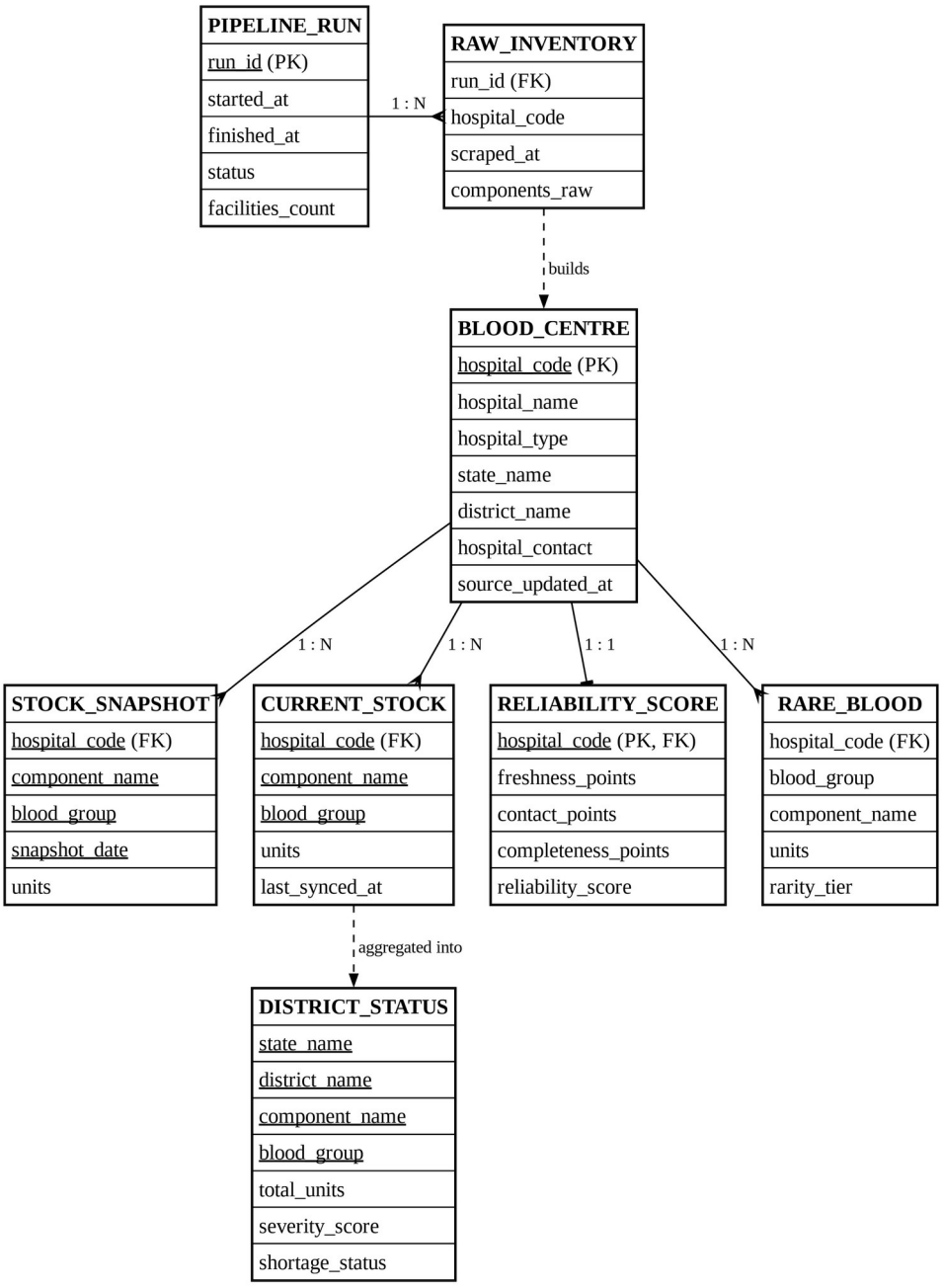


Figure 11: ER Diagram

Data Dictionary

1. Bronze Layer (Raw Data)

Table 5: raw_inventory (bronze.raw_inventory)

Field Name	Data Type	Description
run_id	VARCHAR	Unique identifier of the pipeline/scrape run
scraped_at	TIMESTAMP	Date-time when the record was scraped
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
hospital_type	VARCHAR	Ownership type (Govt, Private, Charitable)
facility_type	VARCHAR	Facility category (e.g. Blood Bank)
hospital_contact	VARCHAR	Phone, fax and e-mail as published
hospital_address	VARCHAR	Postal address of the blood centre
entry_date_raw	VARCHAR	Last stock update time as text from the source
components_raw	VARCHAR	Stock per component and blood group (JSON)
raw_payload	VARCHAR	Complete original record (JSON) for audit
ingested_at	TIMESTAMP	Date-time when loaded into Bronze

2. Gold Layer: Dimension & Facts

Table 6: dim_blood_banks (main_gold.dim_blood_banks)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
hospital_address	VARCHAR	Postal address of the blood centre
hospital_contact	VARCHAR	Phone, fax and e-mail as published
hospital_type	VARCHAR	Ownership type (Govt, Private, Charitable)
facility_type	VARCHAR	Facility category (e.g. Blood Bank)
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
source_updated_at	TIMESTAMP	Last stock update time reported by the centre
lakehouse_ingested_at	TIMESTAMP	Time of the latest scrape containing the centre

Table 7: fct_stock_snapshot (main_gold.fct_stock_snapshot)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
component_name	VARCHAR	Blood component (e.g. Packed Red Blood Cells)
blood_group	VARCHAR	Normalised blood group (e.g. O-, AB+, Bombay)
units	INTEGER	Units available
source_updated_at	TIMESTAMP	Last stock update time reported by the centre
snapshot_at	TIMESTAMP	Scrape time of the snapshot
snapshot_date	DATE	Calendar date of the daily snapshot

Table 8: fct_stock_current (main_gold.fct_stock_current)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
component_name	VARCHAR	Blood component (e.g. Packed Red Blood Cells)
blood_group	VARCHAR	Normalised blood group (e.g. O-, AB+, Bombay)
units	INTEGER	Units available
source_updated_at	TIMESTAMP	Last stock update time reported by the centre
last_synced_at	TIMESTAMP	Time of the centre's latest scrape

3. Gold Layer: Analytical Marts

Table 9: mart_district_status (main_gold.mart_district_status)

Field Name	Data Type	Description
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
component_name	VARCHAR	Blood component (e.g. Packed Red Blood Cells)
blood_group	VARCHAR	Normalised blood group (e.g. O-, AB+, Bombay)
total_reporting_banks	BIGINT	Number of centres reporting this group
stocked_banks	BIGINT	Centres with at least one unit
banks_with_zero	BIGINT	Centres with zero units
total_units	BIGINT	Total units in the district
min_facility_units	INTEGER	Lowest stock at a single centre
max_facility_units	INTEGER	Highest stock at a single centre
shortage_flag	BOOLEAN	True when total units are 5 or fewer
severity_score	INTEGER	Shortage urgency from 0 to 100
shortage_status	VARCHAR	Critical depletion / High / Moderate / Adequate

Table 10: mart_rare_blood_availability (main_gold.mart_rare_blood_availability)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
hospital_type	VARCHAR	Ownership type (Govt, Private, Charitable)
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
hospital_contact	VARCHAR	Phone, fax and e-mail as published
hospital_address	VARCHAR	Postal address of the blood centre
component_name	VARCHAR	Blood component (e.g. Packed Red Blood Cells)
blood_group	VARCHAR	Normalised blood group (e.g. O-, AB+, Bombay)
units	INTEGER	Units available
rarity_tier	VARCHAR	Rarity class of the blood group
source_updated_at	TIMESTAMP	Last stock update time reported by the centre
last_synced_at	TIMESTAMP	Time of the centre's latest scrape

Table 11: mart_reliability_score (main_gold.mart_reliability_score)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
hospital_type	VARCHAR	Ownership type (Govt, Private, Charitable)
facility_type	VARCHAR	Facility category (e.g. Blood Bank)
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
hospital_contact	VARCHAR	Phone, fax and e-mail as published
hospital_address	VARCHAR	Postal address of the blood centre
source_updated_at	TIMESTAMP	Last stock update time reported by the centre
days_since_update	BIGINT	Days since the centre last updated stock
distinct_components_count	BIGINT	Number of components the centre reports
total_units_in_stock	BIGINT	Total units across all components
freshness_points	INTEGER	Trust points for recent updates (max 50)
contact_points	INTEGER	Trust points for a valid phone (max 25)

completeness_points	INTEGER	Trust points for component coverage (max 25)
reliability_score	INTEGER	Trust Score from 0 to 100
reliability_label	VARCHAR	High (75+), Medium (40–74) or Low
reliability_tier	VARCHAR	Same as label, kept for API compatibility

Table 12: mart_stock_change (main_gold.mart_stock_change)

Field Name	Data Type	Description
hospital_code	BIGINT	Unique eRaktKosh code of the blood centre
hospital_name	VARCHAR	Name of the blood centre
state_name	VARCHAR	State or Union Territory
district_name	VARCHAR	District of the blood centre
component_name	VARCHAR	Blood component (e.g. Packed Red Blood Cells)
blood_group	VARCHAR	Normalised blood group (e.g. O-, AB+, Bombay)
current_units	INTEGER	Units in the latest snapshot
prev_snapshot_units	INTEGER	Units in the previous snapshot
delta_prev	INTEGER	Change since the previous snapshot
delta_7d	INTEGER	Change over the last 7 snapshots
risk_flag	VARCHAR	Stable, Falling or Depleted
anomaly_flag	VARCHAR	NORMAL or SUDDEN_DROP_TO_ZERO

4. Metadata

Table 13: pipeline_runs (gold.pipeline_runs)

Field Name	Data Type	Description
run_id	VARCHAR	Unique identifier of the pipeline/scrape run
started_at	TIMESTAMP	Run start time
finished_at	TIMESTAMP	Run finish time
status	VARCHAR	RUNNING, SUCCESS, SUCCESS_NO_SCRAPE or FAILED
facilities_count	INTEGER	Blood centres in the published data
tests_passed	BOOLEAN	Whether all quality checks passed
error_message	VARCHAR	Failure reason, if any
created_at	TIMESTAMP	Row creation time

Screen Layouts

Home Page & National Statistics

√ The first screen shows live national figures: total blood centres, centres updated in the last 24 hours and centres flagged for review.

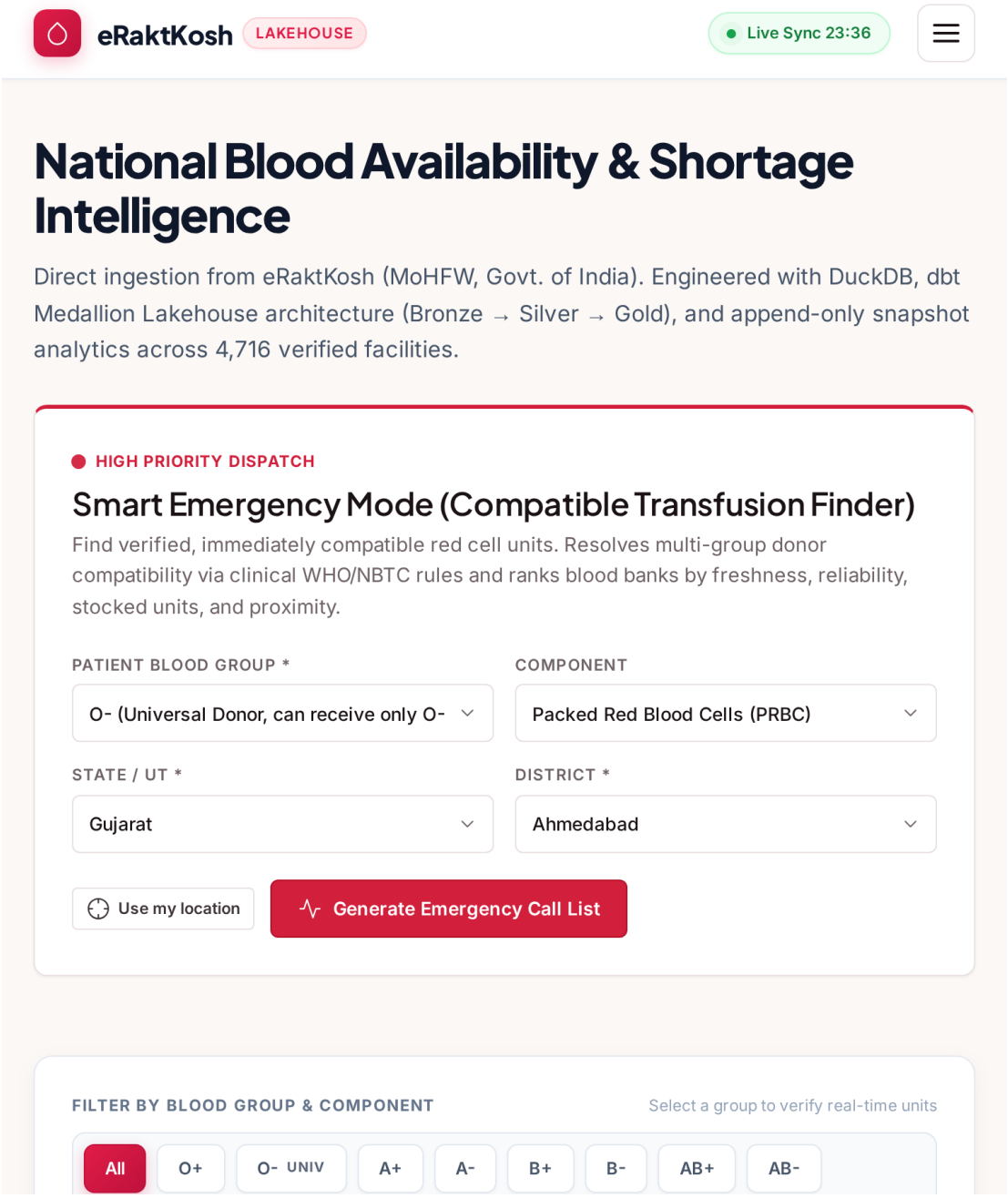


Figure 12: Home Page & National Statistics

Smart Emergency Mode

√ The user picks the patient's blood group, component and place. The system shows safe donor groups, warnings for Rh-negative and Bombay groups, and the best centres with Call and Directions buttons.

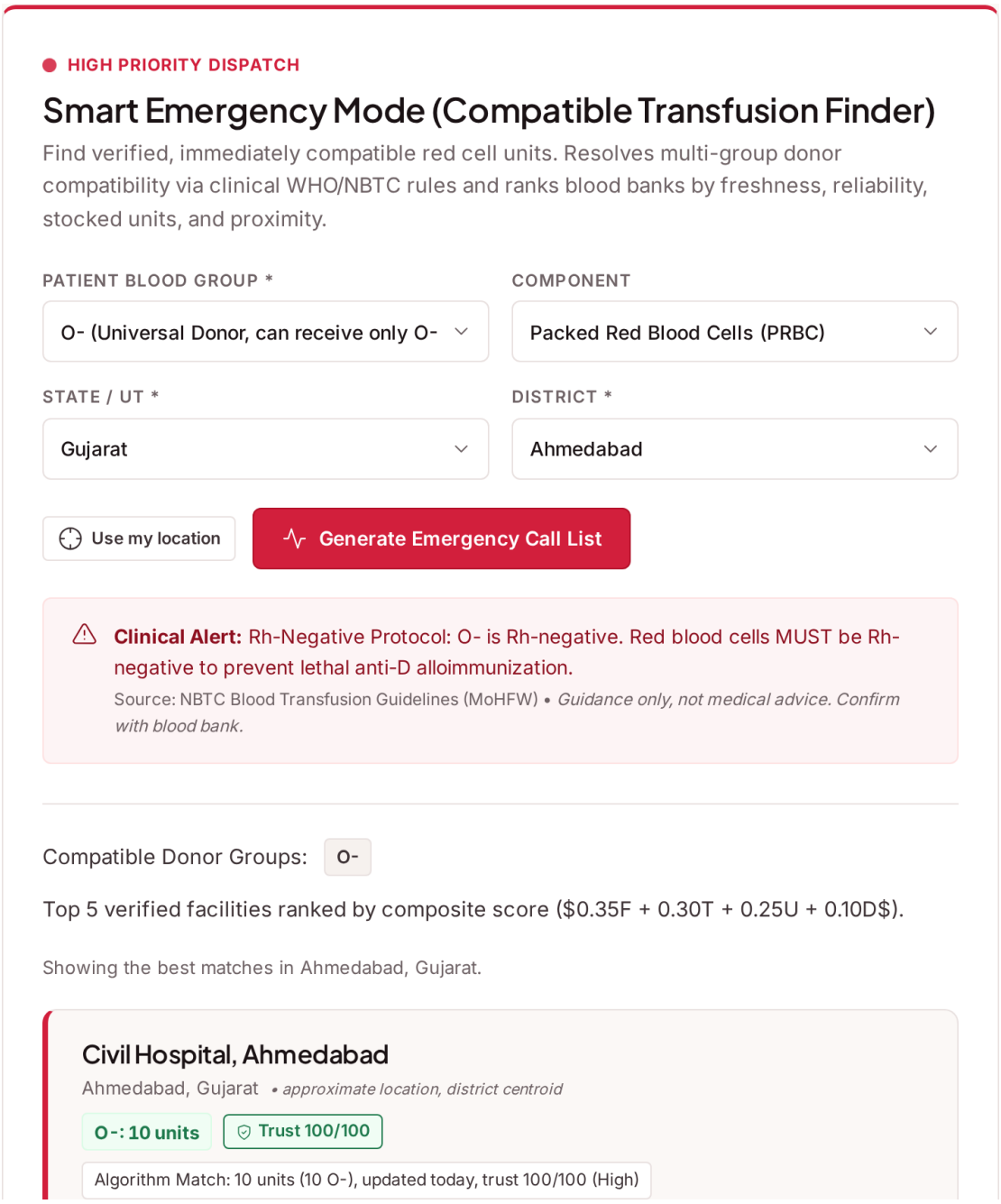


Figure 13: Smart Emergency Mode

Blood Finder

√ Filters for state, district, component, blood group and name, with sorting by freshness, units, trust or distance.

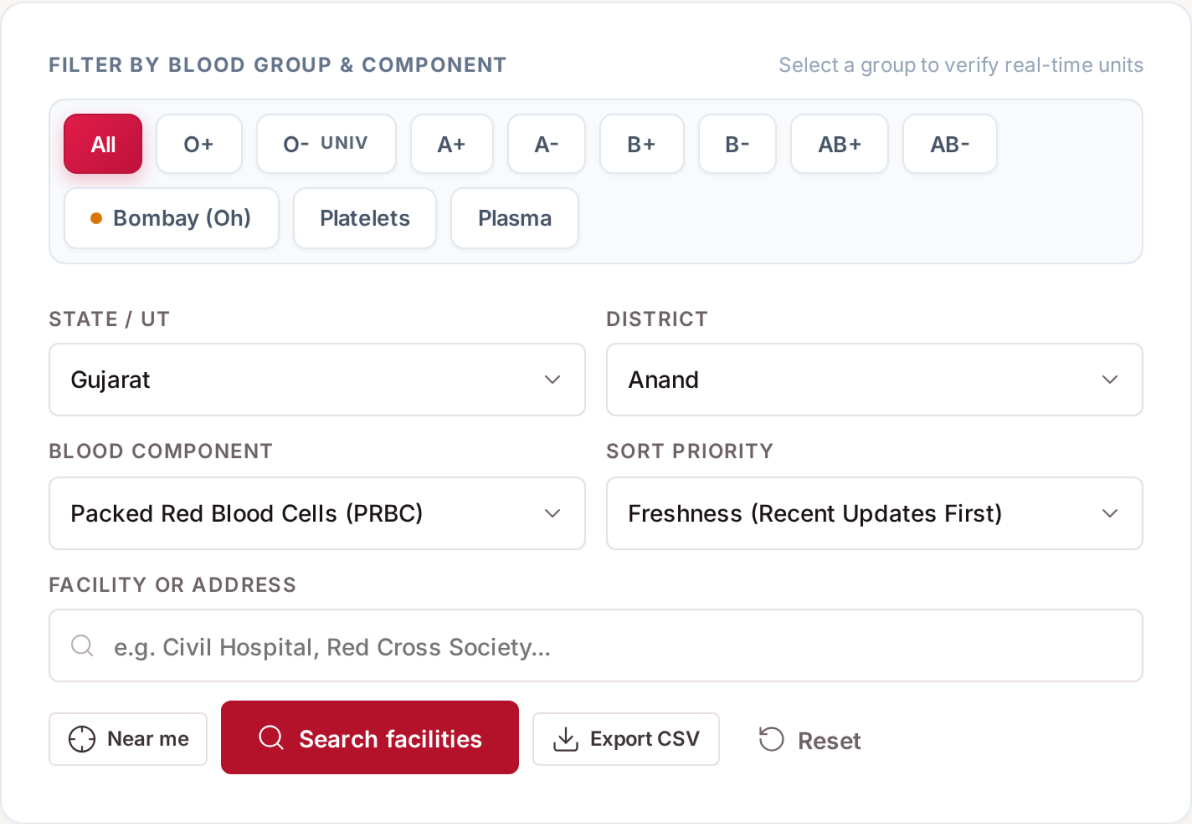


Figure 14: Blood Finder

Blood Finder: Result Cards

√ Each card shows the centre, its location, how recently it updated, its Trust Score and the units for all eight blood groups.

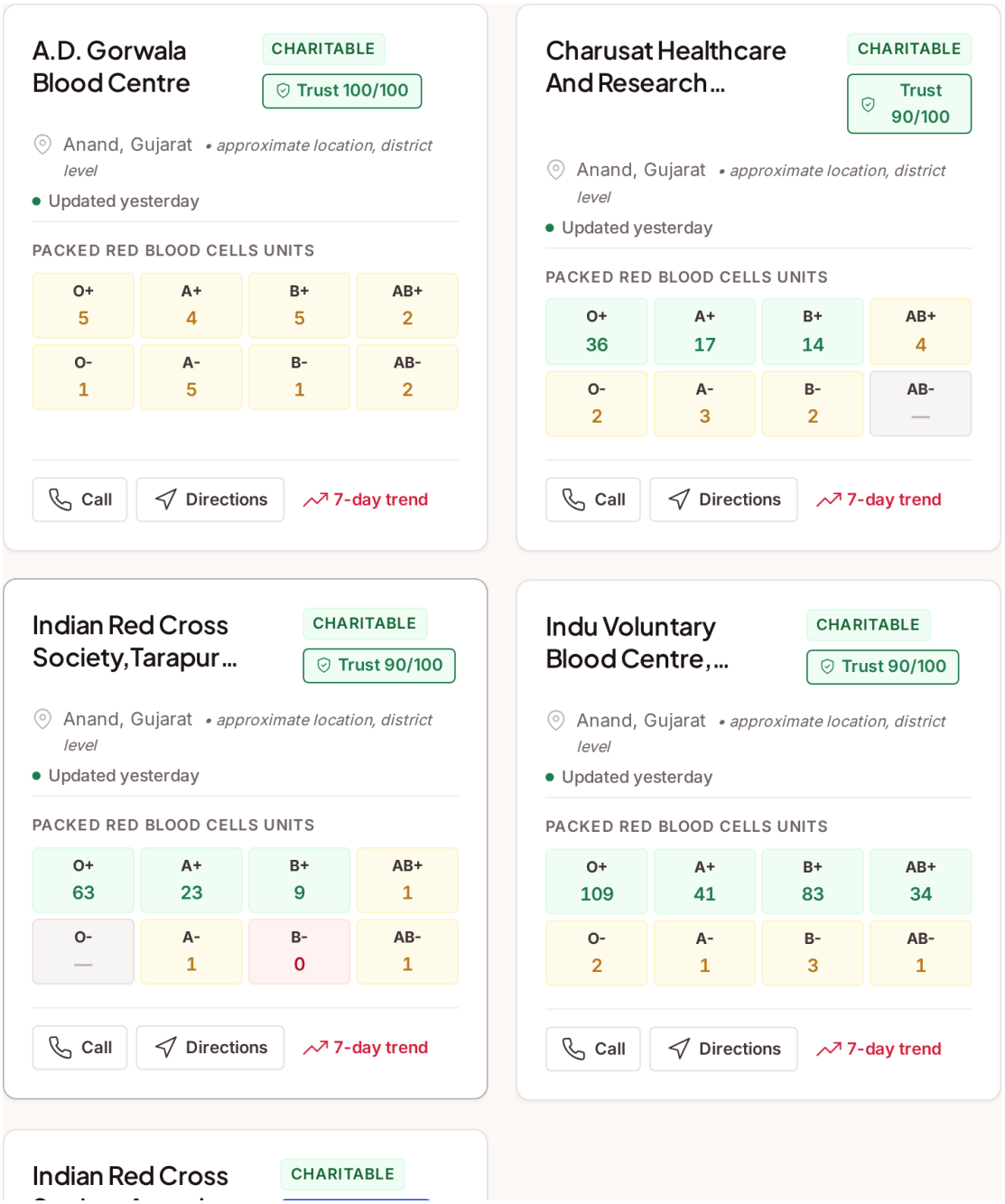


Figure 15: Blood Finder: Result Cards

Live Blood Map

√ Search results on a map. Marker colour shows the stock level; a pop-up offers Call, Directions and the 7-day trend.

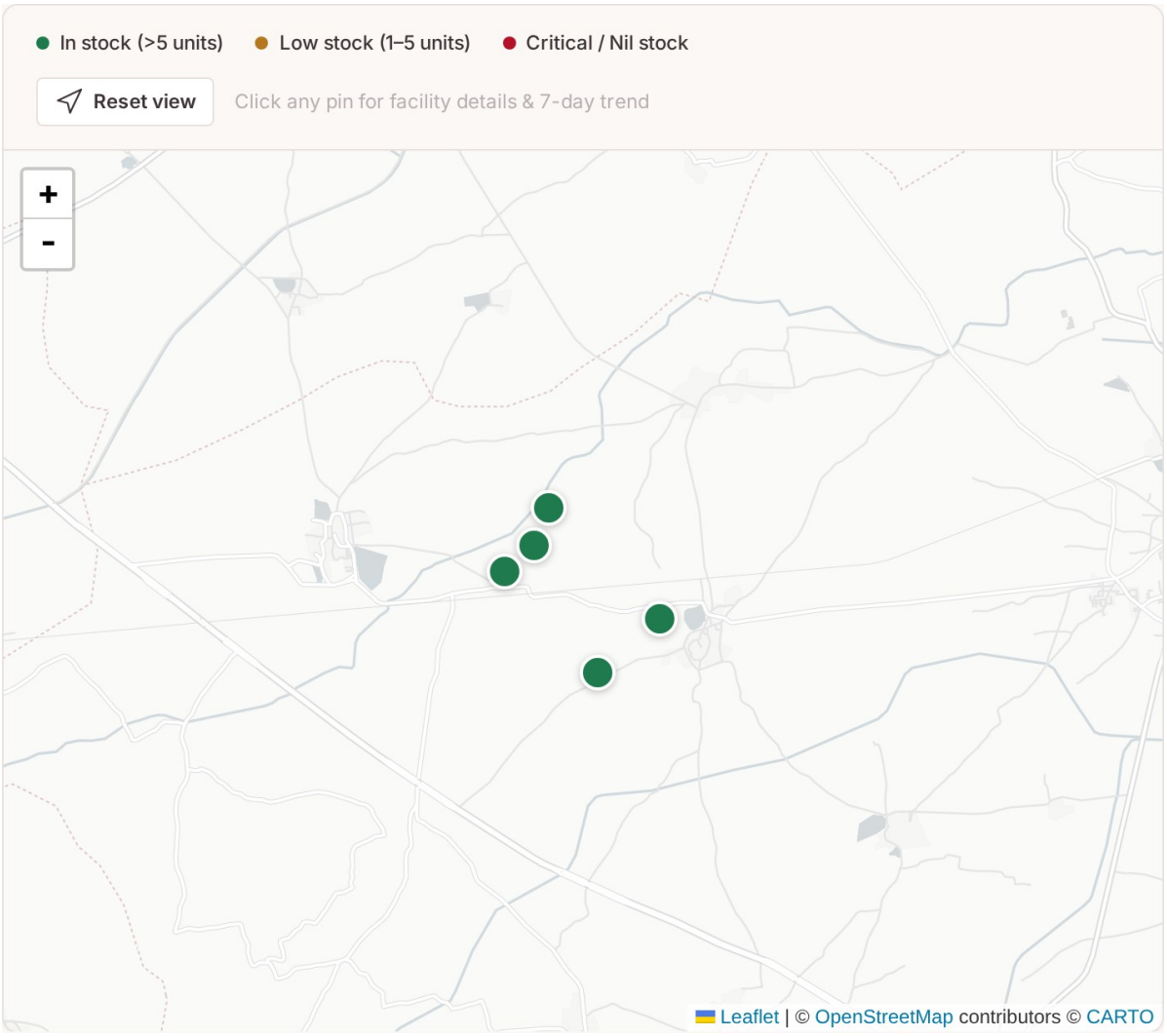


Figure 16: Live Blood Map

7-day Inventory Trend

√ Real daily stock history per blood group, with current units, change and status.

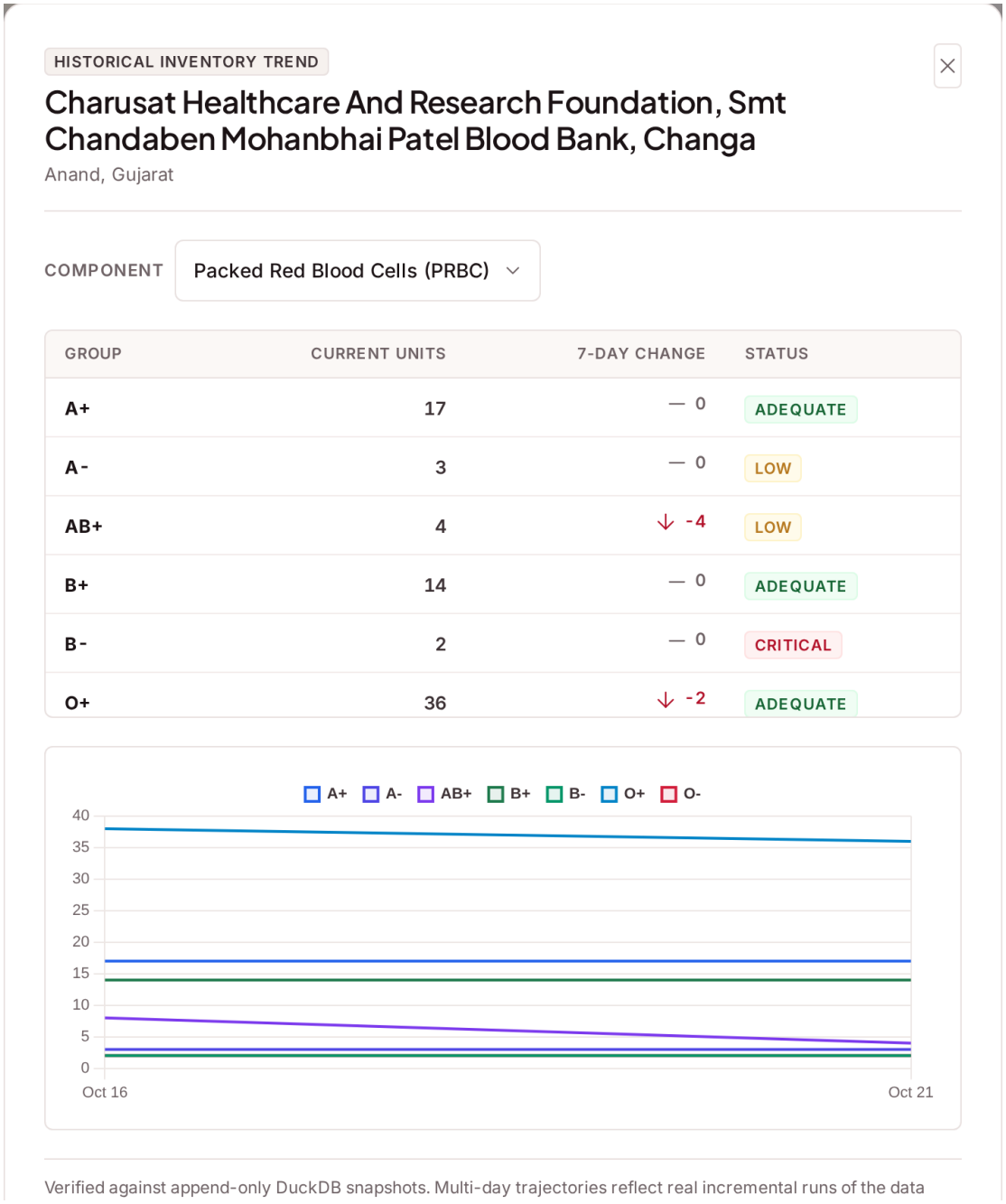


Figure 17: 7-day Inventory Trend

Trust Score Explanation

√ How the 0–100 Trust Score is calculated: freshness (50), valid contact (25) and component coverage (25).

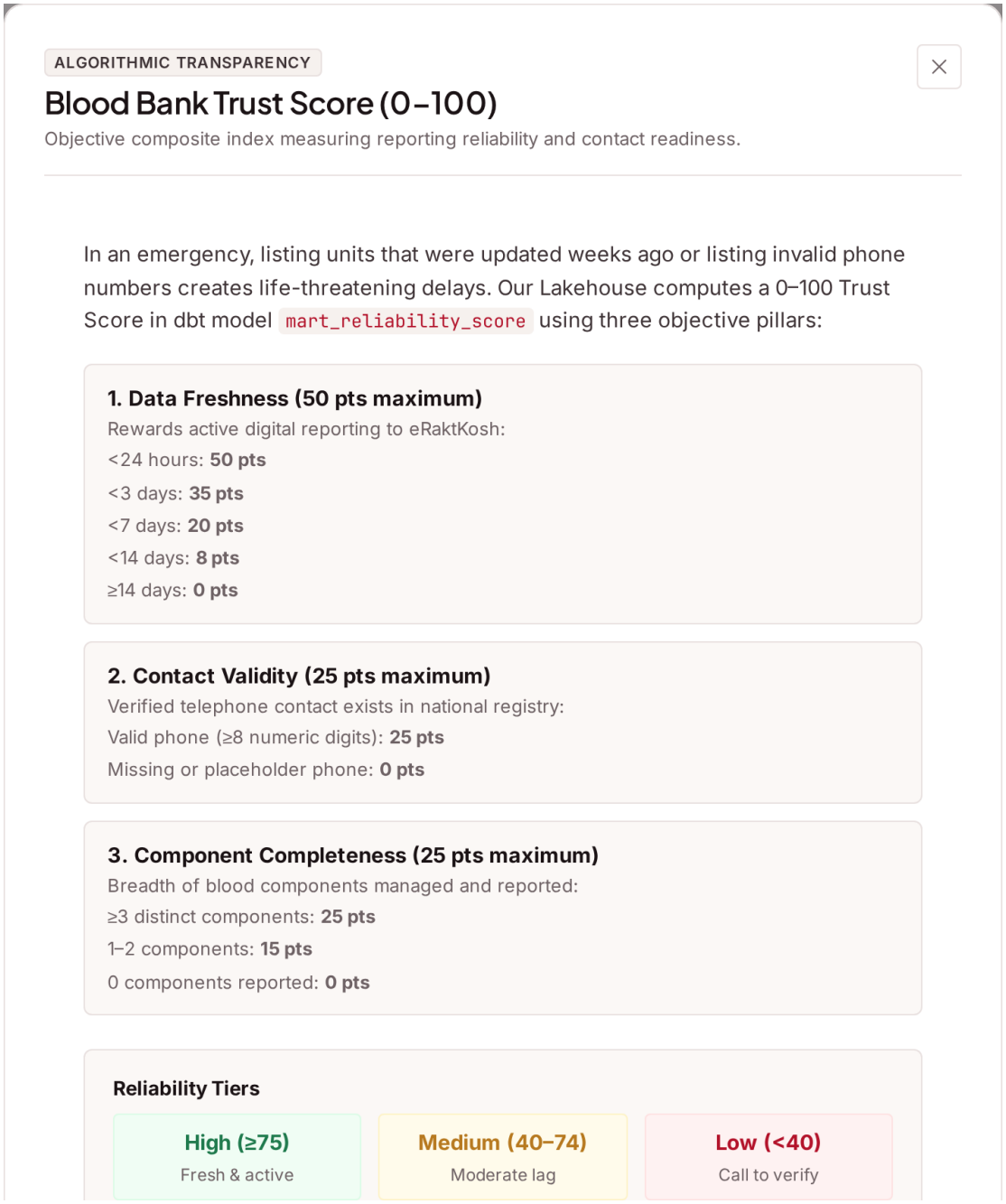


Figure 18: Trust Score Explanation

National Network Distribution

√ Bar chart of active blood centres in the top states.

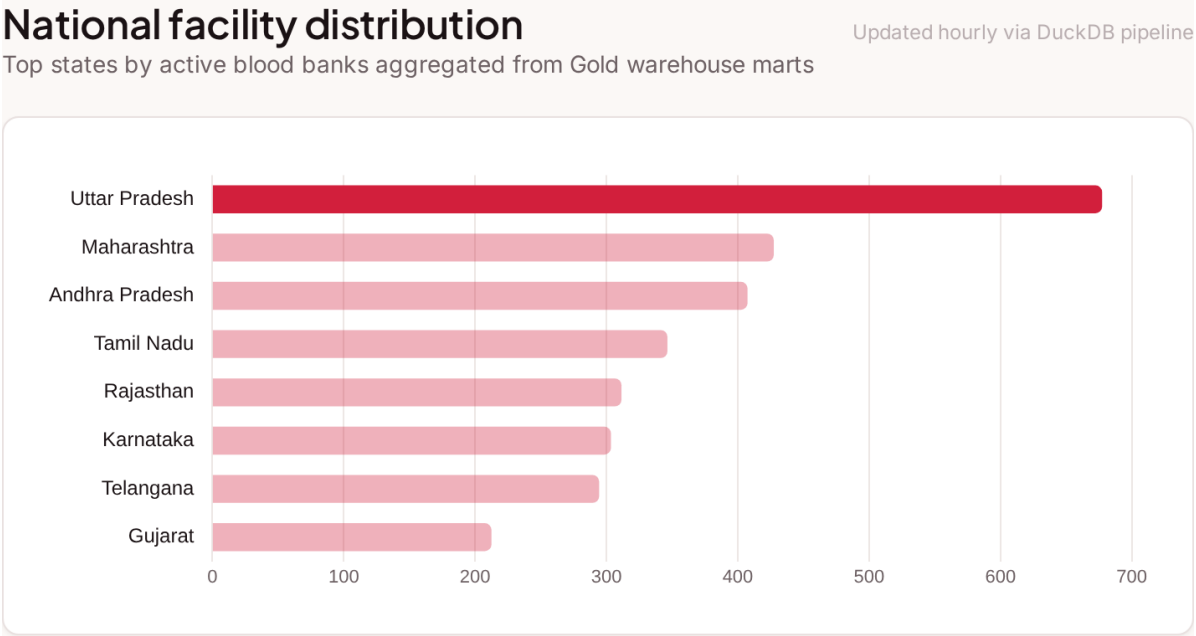


Figure 19: National Network Distribution

Rare Blood Phenotype Radar

√ Centres across India holding Bombay and other rare blood, with contact numbers.

HIGH CLINICAL PRIORITY NATIONAL GOLD MART

Rare Blood Phenotype Radar

Live tracking of Bombay Phenotype (Oh+, Oh-) and high-priority universal negative donor reserves across India.

Sourced from mart_rare_blood_availability

Facility Name	Location	Phenotype	Component	Available Units	Emergency Contact
Rotary Campco Blood Centre	Dakshina Kannada, Karnataka	BOMBAY (OH+)	Fresh Frozen Plasma	676 units	9449733642
Blood Centre Ruhelkhand Medical College	Bareilly, Uttar Pradesh	BOMBAY (OH+)	Packed Red Blood Cells	219 units	7011355320
District Hospital, Jhabua	Jhabua, Madhya Pradesh	BOMBAY (OH+)	Whole Blood	206 units	9855665628
Shree blood centre	Durg, Chhattisgarh	BOMBAY (OH+)	Packed Red Blood Cells	184 units	6263895236
Govt Y.N. (Yagy Naryan) Hospital Kishangarh Ajmer	Ajmer, Rajasthan	BOMBAY (OH+)	Whole Blood	171 units	9649989048
JINDAL HOSPITAL DIAGNOSTICS AND	Seoni,		Fresh	162	

Figure 20: Rare Blood Phenotype Radar

District Shortage Radar

√ Districts running short of the selected blood group, with CSV export for health officials.

SHORTAGE EARLY WARNING AGGREGATED BY DISTRICT

District Deficit Radar

Districts experiencing acute supply depletion (0–2 units) aggregated across all reporting blood centres.

All States

O-

PRBC

Export CSV

District	State	Blood Group	Total Units	Reporting Banks	Banks with 0 Units	Severity Score	Shorta
Anakapalli	Andhra Pradesh	o-	0	2	2	100/100	CRITIC
Kannauj	Uttar Pradesh	o-	0	2	2	100/100	CRITIC
Bundi	Rajasthan	o-	0	1	1	100/100	CRITIC
Singrauli	Madhya Pradesh	o-	0	1	1	100/100	CRITIC
Madhubani	Bihar	o-	0	1	1	100/100	CRITIC
Rohtas	Bihar	o-	0	1	1	100/100	CRITIC
Fazilka	Punjab	o-	0	1	1	100/100	CRITIC

Figure 21: District Shortage Radar

Donor Eligibility & Assistance

√ Checks in the browser whether a person can donate today, using NBTC rules.

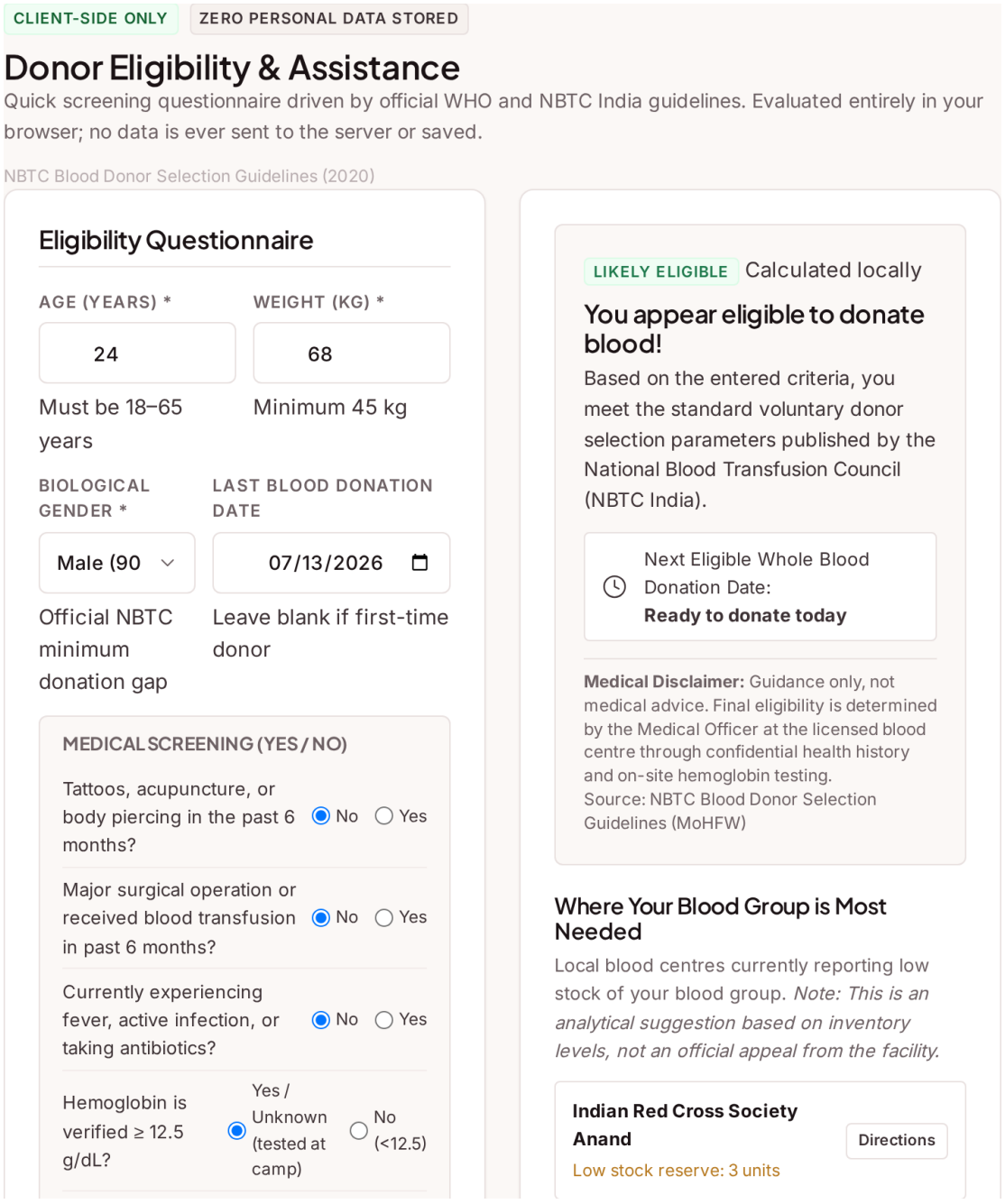


Figure 22: Donor Eligibility & Assistance

Data Health & Pipeline Runs

√ Last successful sync, fresh and stale centres, test results and recent pipeline runs.

LAKEHOUSE OBSERVABILITY DATA QUALITY AUDIT

Lakehouse Data Health & Pipeline Runs

Real-time audit telemetry, pipeline sync age, stale facility distribution, and dbt contract test verification across bronze, silver, and gold lakehouse layers.

Powered by gold.pipeline_runs & DuckDB metadata

Last Successful Sync
2026-10-21 23:36:43
0.8 hours ago

Fresh Facilities (24h)
3,311
70% of total network (4,716 banks)

Stale Facilities (>7d / >30d)
520 / 292
Flagged for registry verification

dbt Schema Quality Tests
37 / 37 PASS
Bronze • Silver • Gold contracts

Top States by Stale Facilities (>7 Days)

State	Total Banks	Stale >7d	Stale >30d
Uttar Pradesh	677	77	48
Karnataka	303	45	37
Kerala	206	41	25
Rajasthan	311	51	22
Tamil Nadu	346	40	20

Figure 23: Data Health & Pipeline Runs

Mobile Responsive Design

√ The same website on a phone: home screen and the menu.

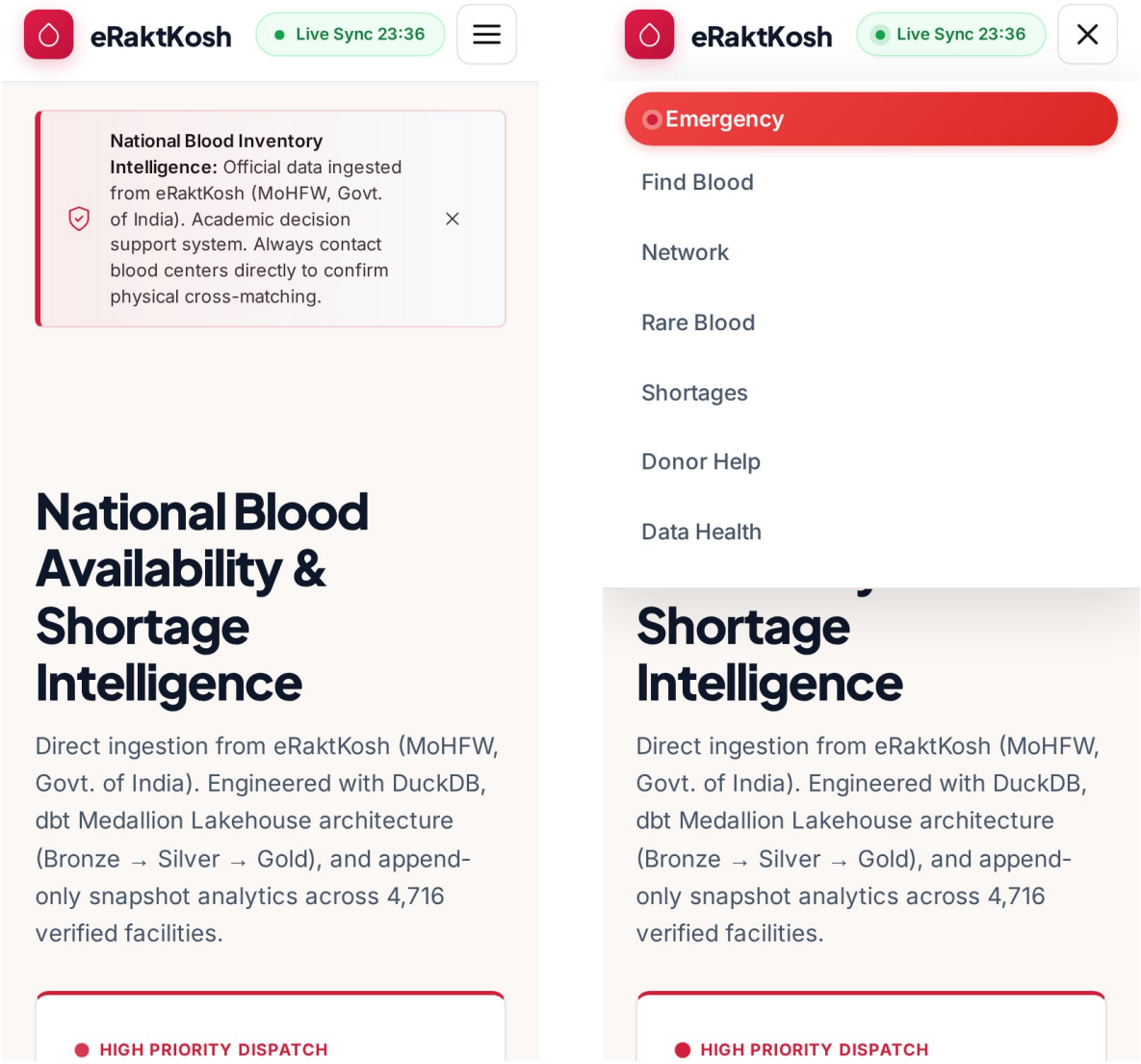


Figure 24: Mobile Responsive Design

Mobile Responsive Design: Search Results

√ Blood centre cards and emergency results on a phone, with large Call and Directions buttons.

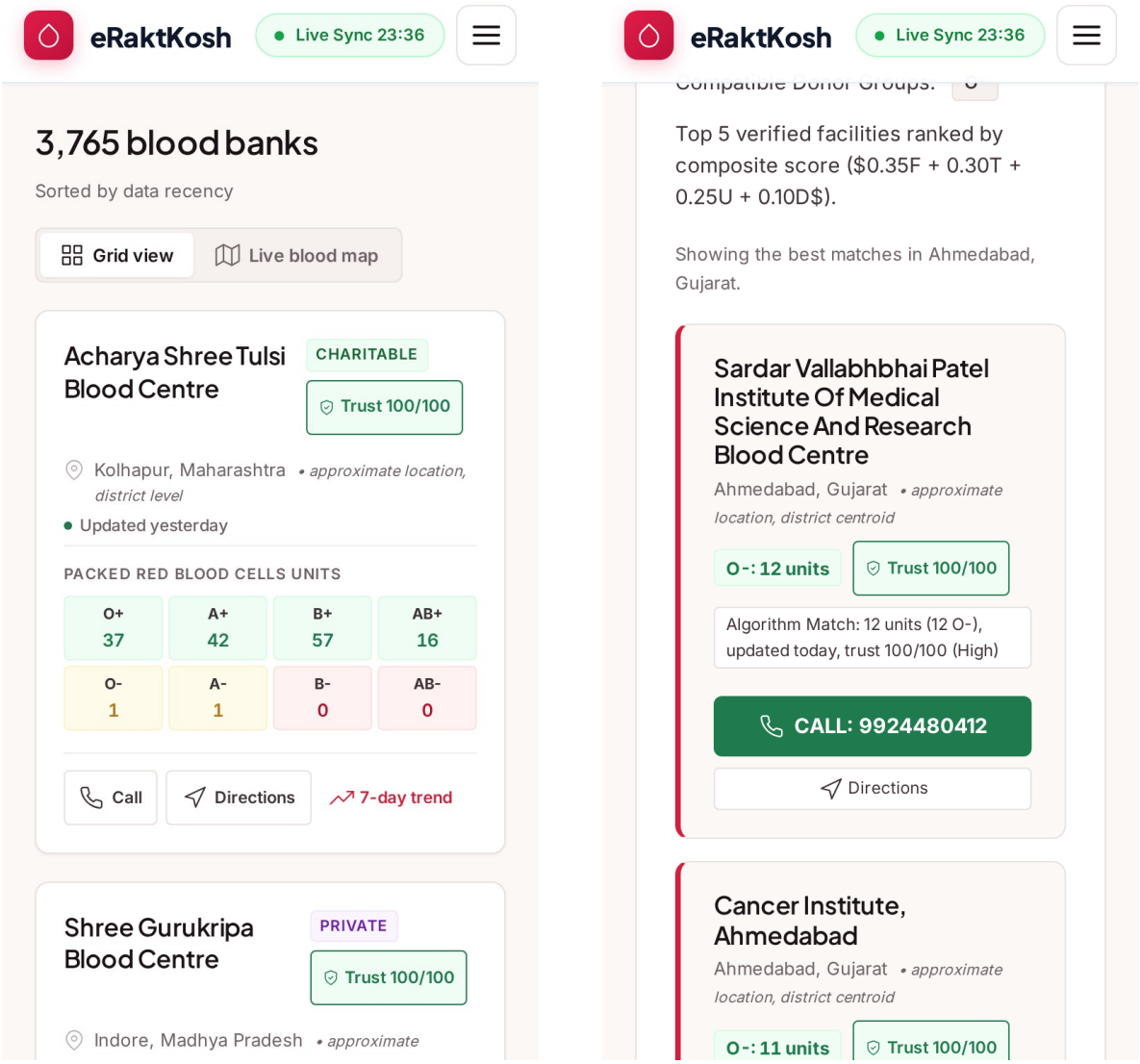


Figure 25: Mobile Responsive Design: Search Results

SYSTEM TESTING

Testing Strategies

√ **Black Box Testing:** Every user flow was tested from the outside by using the real website in a browser (Playwright), on desktop and mobile screen sizes.

√ **White Box (Unit) Testing:** pytest tests check the code inside: blood-group rules, Trust Score, geocoding, CSV format and the scraper safety checks.

√ **Integration Testing:** Every API endpoint is called with FastAPI's TestClient against a test database.

√ **Data Quality Testing:** dbt tests check that keys are unique, required values exist and values are valid in every layer. The anomaly gate checks the facility count.

√ **Responsive Testing:** The layout was checked at 320, 360, 390, 768, 1024, 1280 and 1920 px widths.

√ **Offline (PWA) Testing:** The website was reloaded with the network turned off to check the cache.

Test Results Summary

Table 14: Summary of test results

Test Type	Tool	Number of Checks	Result
Unit & integration tests	pytest	98	All passed
Data quality tests	dbt	37	All passed
End-to-end user flows	Playwright	146	All passed
Responsive layout	Playwright	7 screen widths	No horizontal scrolling
Black box test cases	Manual + automated	28	All passed

Test Cases

Black box test cases: each row was checked on the running website.

Table 15: Black box test cases

Sr. No.	Scenario	Purpose	Input	Expected Output	Status
1	**Home: Load Dashboard**	Verify live statistics load	Open the website	Total, fresh and flagged centre counts shown	✔
2	**Finder: State → District**	Verify dependent dropdowns	Select state Gujarat	District list is filled and enabled	✔
3	**Finder: District Search**	Verify filters apply	Gujarat, Anand, Search	Only Anand centres listed; URL updated	✔
4	**Finder: Blood Group Filter**	Verify group filter	Click B-	Centres with B- stock listed	✔
5	**Finder: No Results**	Verify empty state	Search 'zzzz-no-such-bank'	'No matching blood centres' message	✔
6	**Finder: Reset**	Verify filters reset	Click Reset	All filters cleared; all centres listed	✔
7	**Finder: Sort by Units**	Verify sorting	Select 'Available Units'	Highest stock shown first	✔
8	**Finder: Pagination**	Verify page navigation	Click Next, Last, change page size	Correct page ranges shown	✔
9	**Finder: Near Me**	Verify location ranking	Allow location (Bengaluru)	Bangalore Urban centres about 4 km away first	✔
10	**Card: Call Link**	Verify phone link	Tap Call on a card	Opens dialer with the first phone number only	✔
11	**Card: 7-day Trend**	Verify history modal	Click 7-day trend, change component	Chart and summary table update	✔
12	**Card: Trust Score**	Verify explanation	Click Trust badge	Score method shown; closes with Esc	✔
13	**Map: View & Pop-up**	Verify map view	Switch to map, click marker	Markers and pop-up with actions shown	✔
14	**Emergency: O- Patient**	Verify compatibility	Group O-, Search	Only O- donors; Rh- negative warning shown	✔
15	**Emergency: Bombay**	Verify rare rule	Group Bombay (Oh+)	Only Bombay donors; critical warning shown	✔
16	**Emergency: AB+ Plasma**	Verify plasma rule	AB+, Fresh Frozen Plasma	Only AB donor groups listed	✔
17	**Emergency: Invalid Group**	Verify validation	API call with group 'XYZ'	Error: Unknown blood group	✔
18	**Shortages: Filters**	Verify shortage radar	State Gujarat, group AB-	Only Gujarat AB- districts listed	✔
19	**Export: Availability CSV**	Verify CSV format	Click Export CSV	Valid CSV, header on first row, all rows	✔
20	**Export: Shortages CSV**	Verify full export	Export O- nationwide	All 501 districts exported	✔
21	**Donor: Eligible**	Verify eligibility	Age 25, 70 kg, 100 days	'Likely Eligible' shown	✔
22	**Donor: Under Age**	Verify deferral	Age 17	'Temporarily Deferred' with reason	✔
23	**Donor: Future Date**	Verify date validation	Last donation in future	Clear message to correct the date	✔

24	**Data Health**	Verify monitoring	Open Data Health	Sync age, runs and dbt tests shown	✔️
25	**Mobile Layout**	Verify responsiveness	Open at 320–390 px width	No horizontal scroll; menu button works	✔️
26	**Offline Mode**	Verify PWA cache	Load once, go offline, reload	App opens with offline notice	✔️
27	**Pipeline: Quality Gate**	Verify anomaly gate	Facility count below 80%	Publish blocked; old data stays online	✔️
28	**Pipeline: Repeat Load**	Verify idempotent load	Load the same Parquet twice	Second load skipped; no duplicates	✔️

CONCLUSION

Conclusion

Raktsetu shows that open government data can be turned into a practical emergency tool. The system collects blood stock for 4,716 blood centres across 35 States/UTs, cleans and tests it, keeps a daily history and serves it through a fast website that works on any phone.

Compared with searching the eRaktKosh portal one district at a time, a user can now find compatible blood near them in a few seconds, see how fresh each record is and call the centre with one tap. Health officials get district-level shortage reports that can be downloaded as CSV files.

The project also gave us hands-on experience with modern data engineering: an async scraper, a medallion lakehouse with DuckDB and dbt, automated data-quality gates, a REST API and a responsive Progressive Web App, all backed by automated tests.

Limitations

√ Data is a periodic snapshot of eRaktKosh, not real-time; users must call before travelling.

√ Accuracy depends on blood centres updating their stock on eRaktKosh.

√ Locations are district centres, not exact hospital addresses.

√ History covers only the days the pipeline has run, so long-term trends need more time.

√ 'Near me' works only on HTTPS or localhost because browsers protect location access.

√ **The interface is available in English only.**

√ This is an academic project, not an official government service.

FUTURE ENHANCEMENT

Future Enhancement

❖ Data & Pipeline:

1. Scheduled Scraping

Run the pipeline automatically every 6–12 hours to build a longer daily history.

2. Hospital-level Geocoding

Locate each blood centre from its address instead of the district centre.

3. Time-travel Tables

Apache Iceberg or Delta Lake to query stock as it was on any past date.

4. Advanced Data Quality

Great Expectations checks for unusual values before publishing.

Future Enhancement: Analytics & Machine Learning

1. Stock Forecasting

Predict when a blood group will run out at a centre (e.g. 'O- in about 3 days').

2. Shortage Early Warning

Flag districts trending towards zero, not only those already at zero.

3. Transfer Suggestions

Recommend moving surplus units between nearby centres.

4. Seasonal Analysis

Study stock changes during dengue season, festivals and holidays.

Future Enhancement: User Features

1. Stock Alerts

E-mail or push notification when a needed group becomes available nearby.

2. Blood Centre Portal

Let centres confirm or correct their own stock to raise trust.

3. Donation Camp Calendar

Show camps in districts with shortages.

4. Multilingual Interface

Hindi and Gujarati with medically reviewed wording.

5. Voice Search & Accessibility

Voice input and a full WCAG 2.2 AA audit.

Future Enhancement: Deployment & Operations

1. HTTPS Cloud Hosting

Public deployment so 'Near me' works on every phone.

2. CI/CD

GitHub Actions running all tests and dbt checks on every change.

3. Monitoring & Alerts

Prometheus/Grafana dashboards and failure alerts for the pipeline.

4. Public API Keys

Rate-limited API access for hospitals and research groups.

Long-Term Expansion Opportunities

√ **National Integration:** Partner with state blood transfusion councils for direct data feeds.

√ **Hospital Information Systems:** Offer the API to hospital software for automatic stock lookups.

√ **Inter-state Coordination:** Shortage maps to support movement of blood between states.

√ **Research Data:** Anonymised historical datasets for public-health research.

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APPENDIX

Appendix A: Glossary

Table 16: Glossary of terms

Term	Meaning
API	Application Programming Interface: a way for programs to request data
Bronze / Silver / Gold	Raw, cleaned and analysis-ready layers of the lakehouse
Bombay phenotype (Oh)	A very rare blood group that can only receive Bombay blood
CSV	Comma-Separated Values: a simple table file that opens in Excel
dbt	Data Build Tool: turns SQL files into tested tables
DFD	Data Flow Diagram
DuckDB	A fast analytical database stored in a single file
ELT	Extract, Load, Transform: the order in which the pipeline handles data
FFP	Fresh Frozen Plasma
Haversine formula	Formula for the distance between two points on Earth
Lakehouse	Data design that keeps raw files and clean tables together
NBTC	National Blood Transfusion Council, Govt. of India
Parquet	A compressed, column-based file format for data
PRBC	Packed Red Blood Cells
PWA	Progressive Web App: a website that can be installed and used offline
Rh-negative	Blood without the RhD antigen (e.g. O-, A-)
Trust Score	Our 0–100 score of how reliable a centre's data is
UML	Unified Modeling Language, used for the diagrams

Appendix B: User Manual

❖ For Public Users

1. Emergency search

Open the website, choose the patient's blood group, component, state and district, then press the emergency search button. Call the top centre to confirm.

2. Near me

Press 'Use my location' or 'Near me' and allow location access to sort by distance.

3. Find blood

In Find Blood, choose filters and press Search. Switch to the map view if needed.

4. Trend and trust

On any card, press '7-day trend' or the Trust badge for details.

5. Reports

Press 'Export CSV' in Find Blood or Shortages to download a report.

6. Donor check

In Donor Help, fill in age, weight and the questions, then press Evaluate.

❖ For the Operator

1. Install

Install Python 3.12 and uv, then run: make install

2. Refresh data

Run: make pipeline (scrape, load, transform, test and publish)

3. Start the website

Run: make run, then open http://localhost:8000

4. Check data health

Open the Data Health section or run: make status

5. Rebuild this report

Run: make report

THANK YOU